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**Feature-Based and Object-Based Limits in Visual Working Memory: the Nature of
Contralateral Delay Activity**

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Abstract

Working memory capacity is limited by the number of objects and the number of features per object, but there is conflicting evidence regarding just what is indexed by contralateral delay activity (CDA), a physiological measure of capacity. Wilson et al. (2012) suggested that it is the number of features that primarily matter for the CDA, whereas Quak et al. (2018) suggested that it is only the number of objects that matter, though they did not carry out feature-by-feature analyses that might modify that view. Finally, Woodman and Vogel (2008) found that the primary governing factor of the CDA was the number of objects, yet that it did matter whether the participant was to attend to one particular feature (color) versus all features. Experiment 1 asks whether the number of features matter, in a manner that replicates the situation of Wilson et al. but with better control over potentially confounding perceptual factors. Experiment 2 uses stimuli resembling all three aforementioned studies, with color and stripe orientation as features of one or two objects. Our results resemble those of Woodman and Vogel, although sensitivity to attention to one feature (color) versus multiple features only emerged for a 1-object load. We propose one possible account, an *Object-load feature-difficulty hypothesis*. In it, the number of objects is the primary factor for the CDA, the CDA response to features is not additive, and the CDA sometimes reflects the difficulty of the most challenging of attended features.

Keywords: visual working memory; contralateral delay activity; event-related potentials
object-based storage; feature-based storage

Feature-Based and Object-Based Limits in Visual Working Memory: the Nature of Contralateral Delay Activity

Working memory refers to a small amount of information that can be kept in a readily accessible state temporarily (Cowan et al., 2024). A key question is whether working memory capacity is constrained by the number of objects maintained in this state, the number of features that can be stored, or both (Hardman & Cowan, 2015; Luck & Vogel, 1997; Luria & Vogel, 2011; Quak et al., 2018; Wilson et al., 2012; Woodman & Vogel, 2008). The present study examines this question using the *contralateral delay activity* (CDA), an electrophysiological measure that has been widely used to investigate the dynamic processes of visual working memory. The CDA is a sustained negative waveform observed over posterior scalp sites contralateral to the visual field where memory items are presented. For instance, if a participant is trying to remember objects on the left side of the screen, the CDA will occur as a negative-going deflection in the electrode sites over right visual cortex compared to the left visual cortex and will be sustained throughout the retention interval. As the number of items increases, CDA amplitude also increases until it reaches a point at which it plateaus, reflecting the limited capacity of working memory (Vogel & Machizawa, 2004).

Although the CDA literature is often described as favoring an object-based account, a closer look suggests a discrepancy in that previous studies fall along a spectrum ranging from more feature-based to more object-based outcomes (e.g., Luria et al., 2016; Luria & Vogel, 2011; Quak et al., 2018; Woodman & Vogel, 2008). At one end of this spectrum, Wilson et al. (2012) found evidence that CDA amplitude was modulated by the total number of features, not by the number of objects alone, supporting feature-based accounts. At the other end, Quak et al. (2018)

reported that CDA amplitude primarily tracked object load, supporting object-based accounts. Between these endpoints, Woodman and Vogel (2008) reported a more intermediate pattern: the CDA was dominated by object load; however, they also found feature-based effects in that attending to color produced a smaller CDA than attending to all features. Thus, existing CDA findings are best characterized as a continuum of results rather than as a settled consensus.

The variability in results of the CDA raises an important, unresolved issue. If prior CDA studies differ in the degree to which feature load affects the CDA, what determines where a given study falls on this spectrum? Given that the CDA offers an important perspective on the neural nature of working memory, it is important to understand what the CDA actually reflects and how it relates to behavioral accuracy. The present study aims to resolve two separate discrepancies in the CDA literature. The first is whether one obtains a CDA magnitude that is consistently larger when more features are to be attended (Wilson et al., 2012) or not (Luria & Vogel, 2011; Quak et al., 2018; Woodman & Vogel, 2008). The second discrepancy is whether multifeatured objects give rise to the same CDA processing no matter whether one or all features are to be attended (Quak et al., 2018) or whether one can discern that the CDA is smaller when only a single feature is to be attended (Woodman & Vogel, 2008). We address these questions across two experiments after explaining each question in a little more detail below.

Do Features Add Up in the CDA?

Wilson et al. (2012), as noted, suggested that attended features additively contribute to the magnitude of the CDA. Their participants completed three conditions: a Solo condition (one single-feature item), a Separate condition (three single-feature items presented in distinct spatial locations, one carrying shape, one carrying color, and one carrying orientation to be remembered), and a Bound condition (one three-feature item containing shape, color, and

orientation to be remembered). Notably, both the Separate and Bound conditions required memory for the same three features, but the formats differed: three single-feature objects versus one multi-feature object. Although the overall finding at the peak of the CDA at electrodes PO7/PO8 was statistically tenuous, the graphic results suggested that there was a larger CDA for conditions with three total features (the Separate condition and Bound conditions) compared to when there was just one feature in all (in the Solo Condition). The same was true for an analysis across electrode sites: The Separate versus Bound conditions did not differ. The only paired comparison to reach significance was the Solo vs. Bound comparison, suggesting a special importance of feature load.

Although Wilson et al.'s (2012) findings were intriguing, their design also introduced potential perceptual confounds. Important, the three display types differed not only in the number of remembered features but also in overall visual complexity and the number of spatial locations occupied by the stimuli. Specifically, the Bound and Separate conditions were perceptually richer than the Solo condition, making it difficult to determine whether CDA differences reflected mnemonic load or simply perceptual demands or attentional demands such as distraction from the irrelevant features (e.g., Berggren & Eimer, 2016; Feldmann-Wüstefeld & Vogel, 2019; Gaspar et al., 2016; Gaspelin & Cowan, 2025).¹ In addition, the Separated condition contained more objects than the Bound or Solo conditions. Although some of these imbalances may have cancelled out in the lateralized CDA waveforms, it is plausible that differences in the overall magnitude of the visually evoked ERP contributed to differences between conditions. As a result,

¹ There has been some debate as to how exactly spatial attention contributes to the CDA component. Although Berggren and Eimer (2016) report the CDA may reflect attentional activation, there is also much evidence that this may have occurred due to a strategy whereby participants encoded sequential displays as distinct episodes to perform the task (Feldmann-Wüstefeld et., 2018). Regarding the present study, it suffices to say that the perceptual and attentional demands varied strongly between task conditions in Wilson et al. (2012) making the differences in the CDA difficult to straightforwardly interpret.

it is not entirely clear if differences in the CDA were due to differences in working memory processes, per se, or were instead due to imbalances in low-level sensory energy (the *Hillyard principle*; Luck, 2014).

In sum, although other studies of the CDA have not found that attended features additively contribute to the CDA (Luria & Vogel, 2011; Quak et al., 2018; Woodman & Vogel, 2008), no previous study attempted to ascertain why the attended-feature effect was found in Wilson et al. (2012) by removing the confounding issues and systematically investigating whether a feature load effect remains. That was the purpose of our Experiment 1.

Does Attending to a Single Feature versus All Features Matter?

Woodman and Vogel (2008) found evidence that the CDA responded primarily to the number of objects, but that attention to one versus several features mattered too. They presented colored bars in one experiment in which the bars could differ in orientation, and colored shapes in another experiment. Participants were to pay attention to the bars, the shapes, the colors, or both features. The CDA did not add up across attended features. However, the CDA was markedly smaller for attending to color than for attending to both features, no matter whether the other feature was orientation or shape. The CDA for attending to two features was at a level comparable to that for the more difficult feature. This pattern did, nevertheless, allow the conclusion that paying attention to a single feature (color) versus multiple features made a difference. A result consistent with this finding was obtained by Luria and Vogel (2011) but their stimuli did not allow a comparison of identical stimuli with different attention demands the way Woodman and Vogel's study did.

In contrast, the study of Quak et al. (2018) did not yield the same result as Woodman and Vogel (2008). Quak et al. included 1 or 4 objects per side to be attended, and each object had 1 or

4 features (shape of the rectangle, its internal color, orientation of a black stripe or stripes, and the thickness and numerosity of black stripes within the rectangle (i.e., its spatial frequency). They found that the CDA depended on the number of objects and was comparable for a particular number of objects no matter whether the participant was to attend to one or all features. There was no difference in attention to one versus all features. It is not clear whether the results of Quak et al. could be consistent with those of Woodman and Vogel because they did not break down the results according to different features; a special status for color could be obscured within the conglomerated results.

It should be noted that the issue of object versus feature representation seems largely settled for behavioral results. In these studies, there is clear evidence that the most important factor is the number of objects, but with modification of the level of performance based on the number of features per object to be remembered (Cowan et al., 2013; Forsberg et al., 2023; Hardman & Cowan, 2015; Oberauer & Eichenberger, 2013; Quak et al., 2018). Quak et al. (2018) in particular showed that feature load and object load can both reduce accuracy even when CDA amplitude remains primarily sensitive to object load. Thus, the issue is the nature of the CDA representation, which may reflect processes occurring in the maintenance period without tapping additional processes that affect behavior during encoding or retrieval.

In sum, there have also been conflicting results about whether the attended number of features will influence the CDA component, and how this relates to specific feature types (Quak et al., 2018; Woodman & Vogel, 2008). Problematically, prior studies have had major differences in stimulus configuration and analysis approach (e.g., electrode sites, time windows, etc.) making it unclear why opposing results remain. Our second experiment examines this issue. We use stimuli that are consistent with those of Wilson et al. (2012) but with only color and stripe

orientation dimensions, which also is a kind of condition similar to all three of the most relevant studies: Wilson et al., Woodman and Vogel, and Quak. However, we look for differences between features as only Woodman and Vogel have done. Part of the difficulty in comparing studies is that they have differed not only in their theoretical framing, but also in their task structure, stimulus displays, and CDA measurement choices, including electrode selection and time-window definitions, and we attend to those factors in our study.

The Current Study

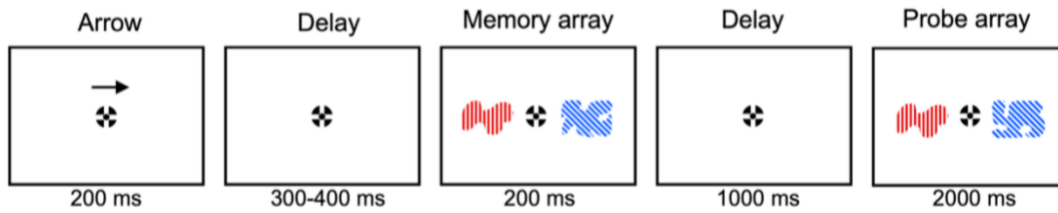
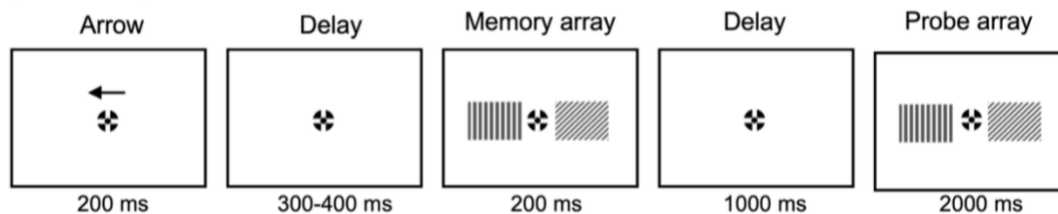
Across two experiments, we aimed to evaluate the contributions of feature load and object load on behavioral performance and the CDA component. Experiment 1 used a task similar to the one used by Wilson et al. (2012) but held the number of objects constant and manipulated the number of task-relevant features using physically identical stimuli. This eliminated perceptual confounds and allowed a more direct test of whether CDA amplitude scales with feature load independent of visual differences. Experiment 2 extended this approach by independently manipulating the number of objects and the number of features in a simplified design that more closely resembled that of Quak et al. (2018), thereby allowing a more direct comparison with a study that reached a strongly object-based conclusion. It allowed us to examine whether a design broadly similar to Quak et al.'s would still yield a purely object-based pattern once feature-specific comparisons were examined more closely.

The present work contributes to the literature in two crucial ways: it provides a more controlled test of Wilson et al.'s (2012) feature-based interpretation, and it examines whether CDA sensitivity to feature load is feature-specific rather than uniform across all probed features. As will be seen, we provide evidence that replicates and extends Woodman and Vogel's (2008) intermediate stance to other experimental configurations.

Experiment 1

Experiment 1 aimed to evaluate Wilson et al.'s (2012) finding that CDA amplitude reflects the number of features maintained in working memory. We used the same stimuli and timing parameters of this study but introduced a critical design modification to minimize perceptual and spatial confounds. Specifically, in Wilson et al.'s study, stimuli were presented in either one or three spatial locations, raising the possibility that perceptual differences rather than working memory load contributed to their ERP results (Luck, 2014). The present design therefore kept the physical stimuli identical across conditions and only varied task instructions.

As shown in Figure 1A, participants were cued to attend a visual hemifield (left or right), followed by a memory display containing one object in each hemifield. After a brief delay, participants were shown probe display and asked to report whether it was the same or different. In separate blocks, participants were instructed to either attend a single feature type of the cued object (*Attend-1-feature*) or all three features of the cued object (*Attend-all-features*), with each in separate blocks. Specifically, in the *Attend-1-feature* condition, participants were required to either *Attend-color*, *Attend-shape*, or *Attend-stripe* in separate blocks. Because the displays were identical across all conditions in our modified design, this minimized perceptual confounds and allowed a more direct assessment of whether CDA tracks the number of features bound within a single object. In addition, we included a control condition in which only one feature was presented to be encoded (Figure 1B). This *Solo-stripe condition* contained only stripe orientation without color or shape. As will be explained shortly, this control condition allowed us to assess potential incidental encoding of irrelevant features in the *Attend-stripe* condition.

Figure 1*Task Design in Experiment 1***(A) Attend-color, Attend-shape, Attend-stripe, and Attend-all Conditions****(B) Solo-stripe Condition**

Note. This figure illustrates the sequence of each trial in the Experiment 1. Panel A depicts the Attend-color, Attend-shape, Attend-stripe, and Attend-all-features conditions, where participants encode and retain either color, shape, stripe, or all three features of the attended objects, with only the block instruction varying across conditions. Panel B depicts the Solo-stripe Condition, in which stimuli contain only stripe orientation, without color or shape. Stimuli are not drawn to scale; feature details are enlarged for visibility.

Experiment 1 examined three research questions. First, we asked whether behavioral performance and CDA amplitude would differ when participants were required to remember three features rather than just one (*Question 1a*). If CDA reflects the number of encoded features, the CDA amplitude should be larger in the Attend-all-features condition than the Attend-1-feature conditions (as in Wilson et al., 2012). If CDA primarily indexes the number of objects, however, CDA amplitude should be equivalent across the Attend-all-features and Attend-1-feature conditions (as in Quak et al., 2018).

If there is a reliable difference between attending one and attending three features, a question remains as to the reason for the difference. First, it is possible that the CDA summates activities from multiple features when attention is directed to all of them. Alternatively, it could

be that features differ in CDA amplitude and the CDA for attending all features reflect not a summation but the amplitude of CDA to the most difficult feature. In order to distinguish between these possibilities, a second type of analysis is needed, in which the CDA is examined separately for each feature in Attend-1-feature vs. Attend-all-features conditions. Thus, we asked whether maintaining multiple features within a single object imposes additional demands beyond those associated with any single feature, after accounting for potential differences in feature-specific difficulty (*Question 1b*). This analysis provided a stricter test of whether maintaining multiple features within one object imposes additional demands.

Finally, we examined whether irrelevant features are encoded automatically by comparing a Solo-stripe condition (stripe presented alone) with the Attend-stripe condition, in which the stripe orientation was embedded within a multi-feature object (*Question 1c*). If CDA increases even when participants are instructed to attend only to stripe orientation of a multifeatured object in the Attend-stripe condition compared to the Solo-stripe condition, this would indicate incidental encoding of task-irrelevant features.

Methods

Participants

A sample of 20 students from the University of Missouri participated for payment \$60 for the 3-hour session. This sample size was determined a priori based on effect size from Wilson et al. (2012), who reported an effect of the posterior CDA peak at PO7/PO8 across feature contexts of $f \approx 0.37$. Based on this effect size, a power analysis indicated that 17 participants would be sufficient to obtain 90% power ($\alpha = .05$) to detect a comparable within-subject effect of CDA. To counterbalance potential order effects, we employed a Latin-square method: because there were five experimental conditions (Attend-color, Attend-shape, Attend-stripe, Attend-all-feature, and

Solo-stripe conditions), we created five corresponding versions in which the starting block differed across versions. To ensure that the five experimental versions were equally represented with four participants each, we set a target sample size of 20 participants. The power required to detect CDA differences depends strongly on the expected effect size and the number of clean trials (Ngiam et al., 2021). The sample size of the present study is also similar to previous studies using the CDA component (e.g., Fukuda et al., 2015; Luria & Vogel, 2011; Woodman & Luck, 2003).

All participants had a normal neurological history, normal or corrected-to-normal vision, and normal color vision as assessed by an Ishihara test. Participants were replaced if their ERP data had an artifact rejection rate exceeding 25% due to eye blinks or eye movements (for more detail, see below) or if they were unable to complete the experiment in a timely manner (i.e., more than 4 hours). Eight participants were replaced for this reason. After these exclusions, data from 20 participants were retained for the final analysis (5 men, 15 women, $M_{age} = 22.7$ years). This study was approved for ethical considerations by the Institutional Review Board of the University of Missouri-Columbia.

Apparatus and Stimuli

Stimuli were presented using PsychToolbox for Matlab on Linux (Kleiner et al., 2007) via an Asus VG248QG LED monitor at a distance of approximately 100 cm in a dimly lit room. A photosensor was used to measure the timing delay of the stimulus presentation system (7 ms), and this was subtracted from all latency values.

Stimuli were adapted from Wilson et al. (2012) and originally developed by Barense et al. (2005). All stimuli presented on a white background with a black fixation cross displayed centrally throughout each trial. A black arrow (2° visual angle in width), appearing 2° above

fixation, indicated whether participants should attend to stimuli presented on the left or right side of the screen. Importantly, these stimulus parameters (e.g., item size, arrow dimensions, and distances) were matched to those reported by Wilson et al., ensuring that the present design closely replicated their visual and spatial characteristics and thereby minimized potential confounding influences arising from low-level perceptual differences.

Each display consisted of two objects, which were unfamiliar geometric polygons. Each object was composed of three distinct visual feature types: color, shape, and stripe orientation. There were four potential photometrically isoluminant colors: red (36.4 cd/m^2 , $x = .646$, $y = .331$), blue (37.2 cd/m^2 , $x = .160$, $y = .132$), green (37.4 cd/m^2 , $x = .253$, $y = .544$), and yellow (37.1 cd/m^2 , $x = .419$, $y = .495$). Stripe orientations included vertical, horizontal, 45° left-tilted, and 45° right-tilted. There were 4 possible polygon shapes, which were generated using stimuli obtained from the original authors (Wilson et al., 2012). Each object was approximately 2.4° in width and 1.7° height, and the internal stripe gratings had a spatial frequency of approximately 3.2 cycles per degree (each light or dark stripe subtended about 0.16°). The stimulus appearance was designed to closely resemble those used by Wilson et al., ensuring comparable visual characteristics across studies. The shapes were also designed to have approximately equivalent area to reduce differences in lateralized stimulus energy. The objects were positioned 5° to the left and right of the central fixation cross, identical to Wilson et al.'s study (2012).

A fixation cross (0.5° in diameter) remained visible throughout each trial to help participants maintain fixation. The fixation cross consisted of a crosshair with a dot in the middle, which has been shown to minimize eye movements compared to simpler fixation markers (Thaler et al., 2013). The color, shape, and stripe orientation of each object were selected independently and randomly on every trial, allowing features to repeat across objects or trials.

This randomization prevented participants from relying on simple repetition cues (e.g., assuming that any repeated feature indicated a “different” response). Additionally, single-stripe stimuli consisted solely of stripe orientation presented within a uniform gray square, without variations in color or shape. These conditions allow us to isolate the effect of a single visual feature on CDA amplitude.

Procedure

As shown in Figure 1A, each trial started with a left or right arrow presents for 200 ms, indicating which side of the screen participants should attend to and remember. After a brief interval during which only the fixation cross remained visible, a memory array appeared for 200 ms, containing two objects. There was one object on each side of fixation. These items were either multi-feature stimuli or single-stripe stimuli. Following this, a 1000 ms delay occurred, during which only the fixation cross remained visible. Finally, a probe array appeared for 2000 ms, showing objects again on both sides. Participants determined whether the attended feature(s) of the object on the cued side had changed from those presented in the memory array.

Before each block, participants received instructions on which feature(s) to focus on. They were instructed to remember either a single specified feature (Attend-color condition, Attend-shape condition, Attend-stripe condition, Solo-stripe condition) or all three features (Attend-all-features condition, in which one of the three features would randomly change with equal chance). Each condition included one practice block of 48 trials, followed by one experimental block of 192 trials (similar to Wilson et al., 2012). There were 5 practice blocks and 5 experimental blocks in total. Participants took a short 30-second break every 48 trials during the experimental blocks. They responded by pressing buttons on a handheld gamepad to indicate

whether the objects are the same or different. They were asked to minimize unnecessary body movements and eye movements to reduce ERP artifacts.

Each block began with a 48-trial practice session. To ensure that participants fully understood the task and could produce analyzable EEG data, the experimenter monitored practice performance and repeated the practice block if accuracy was below 60 % or if more than 30% of trials had artifactual voltages (e.g., eye movements), providing further instruction as needed until these criteria were satisfied.

Behavioral analysis

Behavioral data were analyzed using Bayesian statistical methods in R with the BayesFactor package (Morey & Rouder, 2013). Bayesian ANOVAs and t-tests were employed for hypothesis testing because these methods quantify evidence both in favor of and against the null hypothesis, allowing for clearer interpretation regardless of whether an effect is present or absent. Bayesian analyses utilize a Cauchy prior distribution for effect sizes, centered at zero with a scale parameter of 0.707 (Ly et al., 2018; Rouder et al., 2012). This approach provides flexibility in model comparisons by assessing the relative likelihood of models that either include or exclude specific effects.

To interpret results, we followed guidelines recommended by Rouder et al. (2012). A Bayes factor (BF) of 3 or higher as evidence supporting the presence of an effect, whereas a BF of less than 0.33 indicates evidence favoring the null hypothesis. Bayes factors beyond these thresholds provides increasingly stronger evidence for or against an effect. Although our primary conclusions were based on Bayesian analyses, we also reported traditional frequentist statistics using the sjstats package (Lüdtke, 2019) to provide supplementary descriptive context.

Electrophysiological Recording and Analysis

The electroencephalogram (EEG) data were recorded using an active electrode system (Brain Products ActiCHamp) with 32 Ag/AgCl electrodes placed at 27 scalp locations (Fp1, Fp2, F7, F3, Fz, F4, F8, C3, Cz, C4, P9, P7, P5, P3, Pz, P4, P6, P8, P10, PO7, PO3, POz, PO4, PO8, O1, Oz, and O2), along with five additional external electrodes. Two external electrodes were placed on the left and right mastoids and were used for offline referencing by averaging their signals. Electrooculogram (EOG) was measured using electrodes positioned laterally next to the outer corners of both eyes and below the right eye to detect of horizontal eye movements, vertical eye movements, and eye blinks. Impedances for all electrodes were maintained below 15 k Ω throughout the recording session. EEG signals were recorded at a sampling rate of 500 Hz using Brain Vision Recorder software, incorporating an antialiasing cascaded integrator-comb filter with a half-power cutoff at 130 Hz.

Offline EEG data processing and analysis were performed using EEGLAB (Delorme & Makeig, 2004) and ERPLAB (Lopez-Calderon & Luck, 2014) toolboxes. The continuous EEG data were first re-referenced offline to the averaged signal from the two mastoid electrodes. EEG channels were then filtered offline using a noncausal Butterworth high-pass filter with a half-amplitude cutoff frequency of 0.1 Hz and a slope of 12 dB/octave. Next, EEG data were segmented into epochs ranging from 200 ms prior to stimulus onset (baseline period) to 1400 ms post-stimulus onset, consistent with previous ERP studies (Wilson et al., 2012). The 200-ms prestimulus interval served as a baseline correction period.

Artifact rejection was then conducted on epoched data. To identify eye-related artifacts, bipolar vertical and horizontal EOG channels were computed offline. Trials containing eye blinks were rejected, which were detected via a step-filter as abrupt voltage shifts exceeding 80 μ V within any 200-ms interval on the vertical EOG channel. Additionally, any trials where voltage

exceeds $\pm 100 \mu\text{V}$ at any scalp electrode were excluded for excessive noise. Trials with incorrect behavioral responses or excessively slow reaction times (greater than 2000 ms) were also removed from ERP analyses.

Horizontal eye movements were screened at both the trial and participant levels (Woodman & Luck, 2003). First, we computed a bipolar HEOG channel by subtracting the left-from right-HEOG channels for each subject and a step-filter was applied to each trial and rejected any trial containing a voltage shifts exceeding $16 \mu\text{V}$ within any 100-ms interval. This removed trials with horizontal eye movements from further analysis. Second, to detect any residual systematic eye movements toward the cued hemifield, we calculated the participant-level difference in mean bipolar horizontal EOG activity between cue-right and cue-left trials. Participants were retained only if this difference was less than $\pm 3.2 \mu\text{V}$, corresponding to average eye movements of less than $\pm 0.2^\circ$ toward the cue (Lins et al., 1993; Woodman & Luck, 2003). Step-filter parameters were adjusted, when necessary, guided by visual inspection of the EOG waveforms. Ultimately, participants were replaced if more than 25% of trials were rejected due to artifacts or if grand-averaged bipolar HEOG remained above $\pm 3.2 \mu\text{V}$ for cue-left and cue-right. Eight participants were replaced according to above criteria, which is comparable to prior studies using this approach (e.g., Gaspelin & Luck, 2018; Stilwell et al., 2022; Talcott & Gaspelin, 2021).

In the final sample, 16.0% of trials were rejected due to EEG artifacts. After artifact rejection and exclusion of incorrect trials, the average number of valid trials retained for CDA analysis was 145 out of 192 trials per participant in each condition. To reduce Type I errors, electrode sites and time windows for the CDA were selected a priori (see Luck & Gaspelin, 2017). For the CDA component, ERP waveforms were computed by subtracting ipsilateral from

contralateral activity at electrode sites PO7 and PO8, based upon previous studies of the CDA (Berggren & Eimer, 2016; Luria et al., 2016; Wilson et al., 2012). Importantly, PO7/PO8 were also used by Wilson et al. (2012) to show feature-based effects on the CDA component, which this study directly aims to address. In addition, a recent systematic review showed that these electrode sites are among the most commonly used and exhibit some of the largest reported CDA effect sizes across studies (Roy & Faubert, 2023). CDA amplitude was quantified as the mean voltage difference within the predetermined measurement window (200 to 1200 ms after stimulus onset). For visualization purposes only, averaged ERP waveforms presented in figures were additionally smoothed using a low-pass filter (half-amplitude cutoff at 20 Hz, slope of 12 dB/octave). This visualization filter was not applied prior to statistical measurement of ERP amplitudes.

Openness and Transparency

All data, stimulus presentation scripts, and analysis code for the two experiments are available on the Open Science Framework (anonymous link: https://osf.io/j638t/overview?view_only=87fcf1e5381649c685c0463c926b5ab4). The experiments were not preregistered.

Results

Behavioral Accuracy

Descriptive results: As shown in Table 1, accuracy was highest when participants attended to a single feature. Among the Attend-1-feature conditions, performance was greatest in the Attend-color condition ($M = 0.94$, $SD = 0.04$), followed by the Attend-shape condition ($M = 0.91$, $SD = 0.05$) and the Attend-stripe condition ($M = 0.89$, $SD = 0.06$). In contrast, accuracy declined substantially in the pooled Attend-all-features condition ($M = 0.84$, $SD = 0.05$). For

Table 1

Mean Accuracy and Mean CDA Amplitude for Each Probed Feature in the Attend-1-feature and Attend-All-features Conditions in Experiment 1

Probed Feature	Attend-1-feature		Attend-all-features	
	Accuracy	CDA	Accuracy	CDA
Color	0.94(0.04)	-0.57(0.65)	0.91(0.05)	-1.12(1.02)
Shape	0.91(0.05)	-1.03(0.78)	0.89(0.07)	-1.19(0.85)
Stripe	0.89(0.06)	-1.30(0.80)	0.73(0.10)	-1.26(0.97)
Pooled	0.91(0.04)	-0.96(0.65)	0.84(0.05)	-1.19(0.82)

Note. Values in parentheses represent the standard error of the mean for each condition.

completeness, we also report a pooled Attend-1-feature accuracy ($M = 0.91$, $SD = 0.04$), which collapses across the color, shape, and stripe conditions to account for potential differences in feature difficulty. When only a single object was presented, accuracy declined as the number of to-be-remembered features increased. The reduced accuracy in the Attend-all condition generally supports *feature-based accounts* of visual working memory, indicating increased cognitive demands when participants were required to maintain multiple features simultaneously.

Pooled comparison (*Question 1a*): We first asked whether simultaneously remembering three features impairs performance relative to remembering a single feature. Thus, we first averaged accuracy across the Attend-color condition, Attend-shape condition, and Attend-stripe conditions to create an Attend-1 baseline and compared it with the Attend-all condition (see bottom row of Table 1). A paired-samples frequentist and Bayesian t test revealed that accuracy was significantly lower in the Attend-all-features condition than in the pooled Attend-1-feature condition, with decisive Bayesian evidence favoring the alternative hypothesis ($BF_{10} > 10,000$, $t(19) = -7.30$, $p < .001$). This result demonstrates that, even when only a single object is maintained in working memory, requiring memory for multiple features substantially reduces accuracy, consistent with feature-based accounts of visual working memory.

Controlling for feature type (*Question 1b*): Next, we asked whether remembering three features would impair accuracy compared with remembering a single feature when controlling for individual feature type. This provides a stricter test whether increasing the number of features influences behavioral performance because it equates for potential differences in individual feature difficulty.

We therefore recoded the Attend-all condition into three subconditions according to the probed feature type (color, shape, or stripe) so that each could be directly compared with its corresponding Attend-1-feature condition (Table 1). Then, we conducted a 3 (Feature Type: color, shape, stripe) $\times$ 2 (Number of Attended Features: attend-1-feature vs. attend-all-features) repeated-measures frequentist and Bayesian ANOVA on accuracy. Because our goal was to evaluate the unique contribution of each factor, the Bayesian ANOVA compared the full model including both main effects and their interaction to models in which only the effect of interest was omitted.

The analysis revealed a robust main effect of Number of Attended Features ($BF_{10} > 10,000$, $F(1,19) = 53.30$, $p < .001$), reflecting lower accuracy when participants attended to three features than when they attended to one within one object. There was also a strong main effect of Feature Type ($BF_{10} > 10,000$, $F(2,38) = 45.80$, $p < .001$), indicating that overall accuracy differed across feature types. Finally, there was a reliable interaction between Feature Type and Attend Number ($BF_{10} > 10,000$, $F(2,38) = 19.08$, $p < .001$), showing that the cost of attending to three features was different depending on the feature; next we explore this finding in more detail.

Preplanned pairwise t tests further clarified this interaction. For color, the frequentist analysis indicated a significant difference ($t(19) = 2.23$, $p = .038$), but the Bayes factor ($BF_{10} = 1.71$) provided only anecdotal evidence, suggesting that the support for accuracy difference

between Attend-1-feature and Attend-all-features was relatively weak, with lower accuracy when three features had to be remembered. For shape, the Bayesian evidence was weak ($BF_{10} = 0.60$), and the frequentist comparison likewise showed no significant difference ($t(19) = 1.49, p = .151$). By contrast, for stripe, both analyses provided strong converging evidence ($BF_{10} = 4820.93, t(19) = 6.36, p < .001$) that accuracy was substantially reduced when attending to three features compared to one. In summary, these analyses also reveal a strong feature-based contribution to accuracy. Moreover, the cost of attending to all three features as opposed to attending to only one feature was not uniform across feature types, with the largest difference for tests of stripe orientation.

Solo-stripe condition: To test whether participants automatically encoded task-irrelevant features when the required number of attended features was held constant (*Question 1c*), we compared the Solo-stripe condition (present stripe only, $M = 0.92, SD = 0.05$) with the Attend-1-stripe condition (probe stripe for objects with irrelevant color and shape; $M = 0.89, SD = 0.06$). A paired-samples frequentist and Bayesian t test on accuracy revealed slightly lower accuracy in the Attend-1-stripe condition than in the Solo-stripe condition ($t(19) = 2.94, p = .008$), with Bayesian analysis providing moderate evidence for this difference ($BF_{10} = 5.87$).

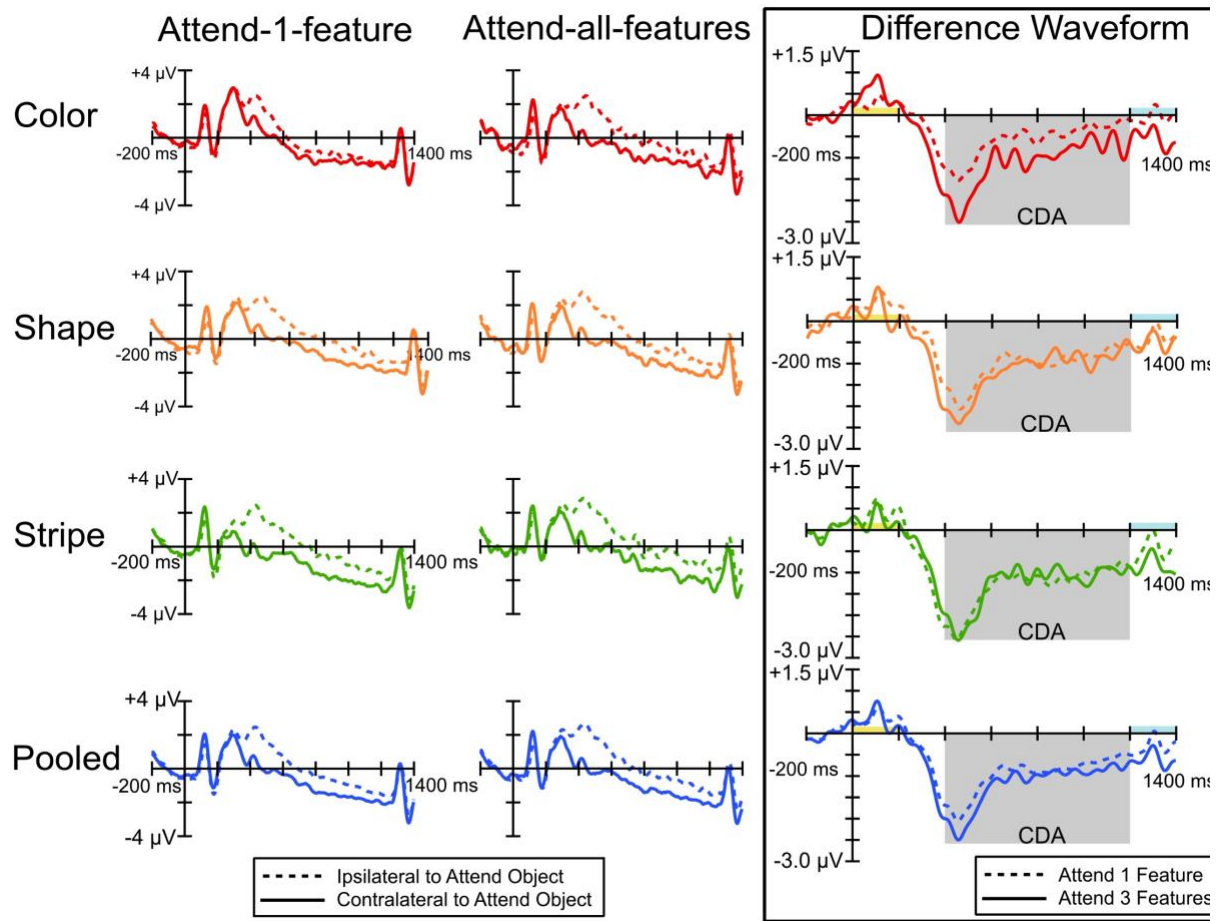
K estimates (items in working memory) for both experiments are reported in the Supplementary Materials (Cowan, 2001; Rouder et al., 2011) and showed patterns similar to those observed for accuracy.

CDA Amplitude

Descriptive results: Figure 2 presents the grand averaged ERP waveforms time-locked to the onset of the memory array from PO7/PO8 electrode sites. For each feature type, the left panels show contralateral and ipsilateral waveforms in the Attend-1-feature condition, and the

Figure 2

ERP Waveforms for Each Attention and Feature Type Condition for multi-featured objects in Experiment 1



Note. For each probe feature type (Color, Shape, Stripe, and Pooled), the left and middle columns display grand averaged ERP waveforms at posterior electrode sites (PO7/PO8) contralateral (solid lines) and ipsilateral (dashed lines) to the attended object are shown separately for Attend-1-feature and Attend-all-features conditions, respectively. The right column shows corresponding difference waveforms (contralateral minus ipsilateral), highlighting the CDA time window (gray shading). For the CDA, the yellow bar on the X axis marks the interval corresponding to the memory array presentation (0-200 ms), and the blue bar on the X axis at the end of the epoch marks the onset of the response period (1200-1400 ms). All waveforms were low-pass filtered at 20 Hz for visualization purposes only and statistical analyses were performed on data without a low-pass filter.

middle panels show the same waveforms in the Attend-all-features condition (for trials in which that feature was probed). The right panels display the corresponding contralateral-minus-ipsilateral difference waveforms, which isolate the CDA component. The shaded gray region highlights the time window (400-1200 ms) used for calculating mean CDA amplitude for each

condition and Table 1 summarizes mean CDA amplitude for each respective condition.

As a general overview, there were no substantive feature load effects. That is, for each feature type, the Attend-1-feature and Attend-all-features conditions produced a similar CDA component. Below, we will be examined the CDA using a series of analyses that are complementary to those above on behavioral performance.

Pooled comparison (*Question 1a*): Mean CDA amplitude did not differ between the pooled Attend-1-feature and Attend-all-features conditions and Bayesian analysis providing only weak evidence regarding the null versus alternative ($BF_{10} = 0.78$, $t(19) = 1.70$, $p = .106$). Thus, although we cannot entirely rule out a contribution of feature-load on the CDA, the feature-load effect was relatively small and nonsignificant, in contrast with the large accuracy differences observed due to feature load (see also Experiment 2).

Controlling for feature type (*Question 1b*): We next examined whether attending to three features, compared with one, modulated the CDA amplitude for each different feature type. As explained above, the CDA might reflect a summation of activities from multiple features in the Attend-all-features condition. Alternatively, it could be that different features differ in CDA amplitude and the CDA for attending all features reflect not a summation but the amplitude of CDA to the dominant feature. This analysis therefore compared Attend-1-feature and Attend-all-features for each feature type (color, shape, stripe orientation).

We conducted a 3 (Feature Type: color, shape, stripe) $\times 2$ (Number of Attended Features: attend-1-feature vs. attend-all-features) repeated-measures frequentist and Bayesian ANOVA on mean CDA amplitude. Bayesian analyses provided strong evidence for a main effect of Feature Type ($BF_{10} = 9.10$, $F(2,38) = 7.55$, $p = .002$), indicating that CDA amplitude differed across feature types. In contrast, the evidence for a main effect of Number of Attended Features was

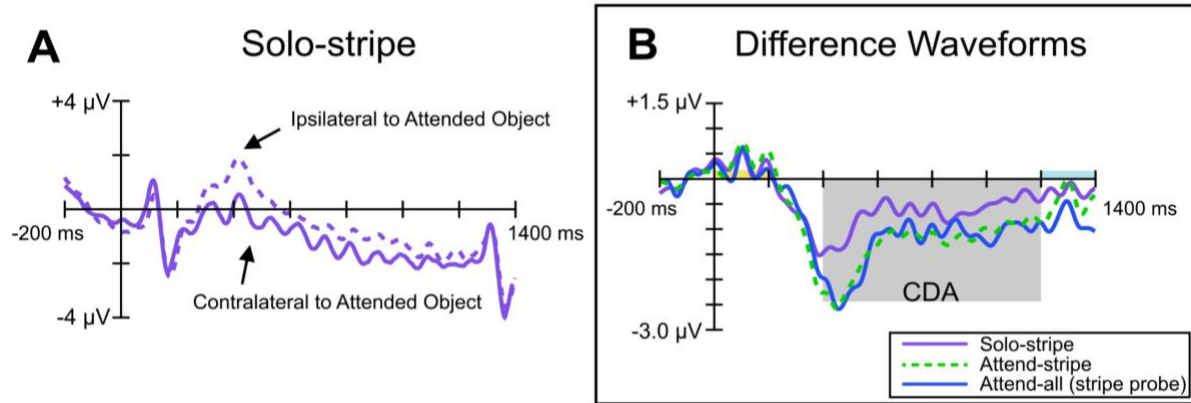
only anecdotal ($BF_{10} = 1.48$, $F(1,19) = 2.59$, $p = .124$), and the evidence for the interaction between Feature Type and Number of Attended Features was also weak, though statistically significant via frequentist statistics ($BF_{10} = 1.04$, $F(2,38) = 3.538$, $p = .039$). These results indicate that CDA differences were driven by the attended feature type (and its associated perceptual or representational demands), with only limited and context-dependent sensitivity to feature load.

To further characterize these patterns, we conducted preplanned pairwise comparisons between the Attend-1-feature and Attend-all-features conditions separately for each feature type (color, shape, stripe). Bayesian results indicated moderate evidence for a difference for color and this pattern was strongly significant via frequentists statistics ($BF_{10} = 4.40$, $t(19) = 2.78$, $p = .012$), but evidence favored the null for shape ($BF_{10} = 0.34$, $t(19) = 0.91$, $p = .373$) and stripe ($BF_{10} = 0.24$, $t(19) = -0.18$, $p = .856$). Thus, the number of attended features did not produce a consistent CDA increase across all three feature types; but an attended feature load effect was observed in the color condition, with little evidence of differences for shape or stripe orientation when only a single object was maintained.

Solo-stripe condition: Figure 3 displays the grand averaged ERP waveforms for Solo-stripe condition ($M = -0.68$, $SD = 0.67$) and CDA difference waveforms for comparing neural activity underlying the Solo-stripe, Attend-stripe, and Attend-all-test-stripe conditions. The CDA emerged between approximately 400 and 1200 ms, with a larger amplitude in the Attend-stripe and Attend-all-test-stripe condition than in the Solo-stripe condition.

Figure 3

ERP Waveforms Comparing of Solo-Stripe, Attend-Stripe, and Attend-all-test-stripe Conditions in Experiment 1



Note. (A) Grand averaged ERP waveforms at posterior electrode sites (PO7/PO8) contralateral (solid line) and ipsilateral (dashed line) to the attended object in the Solo-stripe condition. (B) CDA difference waveforms (contralateral minus ipsilateral) comparing three stripe-related conditions. The shaded gray region indicates the CDA measurement window (400-1200 ms). The yellow bar on the X axis marks the 0-200 ms interval corresponding to the memory array presentation, and the blue bar on the X axis at the end of the epoch marks the onset of the response period.

For the mean CDA amplitude (see Table 1), Bayesian analysis provided strong evidence for a difference between Solo-stripe and Attend-stripe conditions ($BF_{10} = 148.51$), with mean CDA amplitude significantly larger in the Attend-stripe condition than in the Solo-stripe condition ($t(19) = -4.58, p < .001$). These results indicate that even when only stripe orientation was task-relevant, the presence of additional color and shape elicited stronger CDA activity. This pattern suggests that CDA amplitude is sensitive not only to the selected feature per se, but also to the perceptual demands associated with the stimulus and task context.

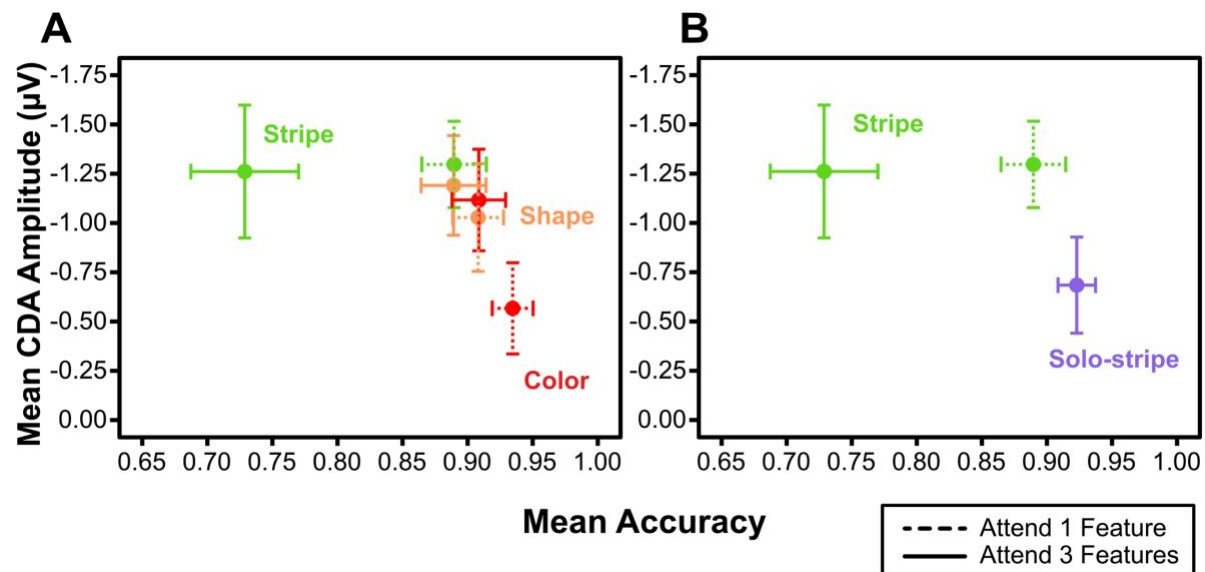
Relationship between Behavioral Accuracy and CDA Amplitude

To visualize the relationship between behavioral performance and electrophysiological activity, we generated plots depicting mean accuracy and mean CDA amplitude across conditions that varied in feature load while holding the number of objects constant (Figure 4A), as well as across conditions differing in the presence of task-irrelevant features (Figure 4B). As shown in

Figure 4A, conditions sharing the same probe feature type (indicated by the same color) differed primarily in the number of attended features. Increasing the number of attended features from one to three was associated with a reduction in behavioral accuracy for stripe and shape conditions. In contrast, CDA amplitude showed relatively little variation within a given feature type across levels of feature load, except for the color condition. Figure 4B shows that CDA amplitude was larger in the Attend-stripe condition than in the Solo-stripe condition, suggesting that the presence of additional, task-irrelevant features contributed to increased CDA activity. Behavioral accuracy could reflect feature-based demands, whereas CDA amplitude primarily indexes the number of maintained objects and is largely insensitive to increased feature load within a single object (although it is small for attention to color alone).

Figure 4

Mean Behavioral Accuracy and CDA Amplitude Across Conditions for Experiment 1



Note. Each point represents a condition average. Error bars denote within-subject 95% CIs (Morey, 2008).

Negative Slow Waveform (NSW)

In the Supplementary Materials, we conducted an exploratory analysis of the Negative Slow Waveform (NSW). The NSW has been previously shown to reflect the bilateral number of items being encoded or maintained in visual working memory (Feldmann-Wüstefeld, 2021; Fukuda et al., 2015;). Cai et al. (2022) further showed that context-binding demands affected bilateral ERPs but not CDA amplitude. These findings motivated our exploratory analysis of whether feature load modulated the NSW. We therefore computed the waveform and compared when it was modulated by feature load as an additional index of working memory. To summarize here, the NSW was calculated as an average of the contralateral and ipsilateral waveforms separately for Attend-one (pooled) and Attend-all conditions. We then computed the mean amplitude of the NSW in the CDA time window. Similar to the CDA, we found no evidence that feature load modulated the NSW, $BF_{10} = 0.33$; $t(19) = -0.88$; $p = .390$. This result provided modest evidence for no effect of feature load on NSW.

Discussion

Experiment 1 examined the findings of Wilson et al. (2012), who reported feature-based limits in visual working memory. Using identical stimulus parameters and timing, we adopted a slightly different design that, we believe, minimized potential sensory confounds. Across a series of focused tests, three main findings could be drawn. First, when the number of objects was fixed at one, behavioral accuracy declined markedly as the number of to-be-remembered features increased. Second, despite this behavioral cost, CDA amplitude was not reliably modulated by the number of attended features when pooled across all feature types. This effect, however, was not consistent across all probed features. A feature-load effect was observed for color probes, whereas the corresponding comparisons for shape and stripe provided little evidence of a difference. Thus, Experiment 1 did not show a general sensitivity of CDA amplitude to instructed

feature load, but instead suggested a more limited, feature-specific pattern. In the context of the broader literature, these results seem more consistent with Woodman and Vogel (2008) than Wilson et al. (2012) or Quak et al. (2018).

One seeming anomaly is that behavior showed relatively little difference between attend-color and attend-all-features responding, whereas the CDA showed a large difference between these conditions. For a possible explanation, assume that information about a single color can be carried by sensory memory with very little contribution of attention. Then even though a participant must work harder to attend all features and this extra effort shows up in the CDA, when the probe turns out to be color it has not been drained much by multifeatured attention because sensory memory of the color is still present.

Although early conceptions of visual sensory (iconic) memory limited it to a fraction of a second, Cowan (1988) proposed that there was a longer, second phase of sensory memory in all modalities that lasted several seconds, yet was not susceptible to severe limitations of the number of items held or the use of attention for maintenance, a description consistent with what was later called fragile visual memory (e.g., Vandembroucke et al., 2011). This longer visual memory might retain the color information in sufficient detail independent of factors that influence the CDA.

A final important result was that, within the Attend-stripe condition, CDA amplitude was larger for the standard multi-feature displays than for the Solo-stripe condition (in which only stripe orientation was shown), indicating that CDA amplitude is sensitive to the broader visual perceptual demands associated with stimulus complexity and task context. This pattern is also consistent with the possibility that some task-irrelevant features are encoded alongside the relevant feature.

Together, these findings suggest that although behavioral performance reflects the amount of feature information maintained, CDA amplitude is much less sensitive to instructed number of features, at least at the global level, and any such sensitivity may be selective to particular feature types rather than uniform across conditions, similar to the conclusions of Woodman and Vogel (2008).

Experiment 2

Experiment 2 aimed to directly compare the number of objects and number of features as independent sources of working memory storage using a simplified design. One possible issue with Experiment 1 is that complex, irregular shapes may have introduced additional perceptual or encoding demands that confounded pure memory effects. Indeed, previous studies have noted that increased object complexity or irregular forms can place additional burdens on encoding and comparison processes, thereby complicating the interpretation of load effects (e.g., Alvarez & Cavanagh, 2004; Gao et al., 2009; Luria et al., 2010). To minimize such influences, Experiment 2 employed simplified stimuli varying only in color and stripe orientation.

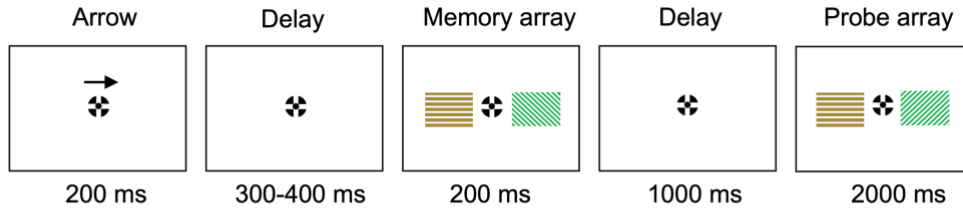
Experiment 2 aimed to address three questions about how object load and feature load are reflected in behavioral performance and CDA amplitude. As shown in Figure 5, the design fully crossed the number of features (*Attend-color*, *Attend-stripe*, *Attend-both-features*) and the number of objects presented (*1 object* vs. *2 objects* per hemifield), yielding six experimental blocks. For example, 1-object, Attend-color means that participants attended only to color, with

one object presented on left side and right side of the fixation.

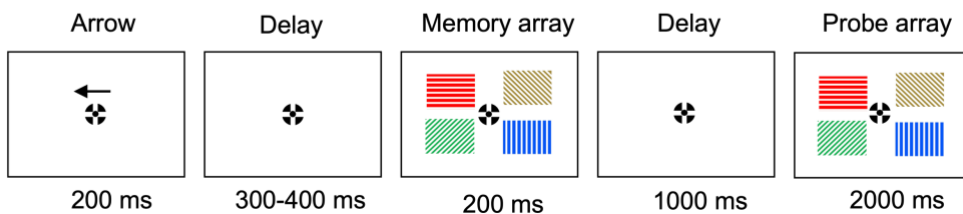
Figure 5

Task Design in Experiment 2

(a) 1-object Condition



(b) 2-object Condition



Note. Panel A is an example of the 1-object condition, in which participants attended to a single object's color, stripe orientation, or both features, depending on block instructions. Panel B an example of the two-object condition, in which participants attended to both objects in the cued hemifield. Across conditions, the number of objects and the number of features to be remembered (one vs. two) varied according to block instructions. Stimuli are not drawn to scale; feature details are enlarged for visibility.

First, we asked whether behavioral performance and CDA amplitude are influenced by the number of objects held in working memory and/or the number of features participants are instructed to maintain (*Question 2a*). Because these factors were manipulated independently, this experiment allowed us to test whether behavior and CDA track the number of objects, the number of features, or both. Second, we asked whether behavioral performance and CDA amplitude scale with the number of objects when the number and type of attended features is held constant (*Question 2b*). If maintaining more objects imposes additional demands, then accuracy should be lower and CDA amplitude should be larger when participants remember two objects than when they remember one object, regardless of the demands on featural attention.

Third, continuing with our more fine-grained analysis, we asked whether increasing feature load by requiring participants to attend multiple features impairs behavioral performance, and whether such costs are reflected in CDA amplitude no matter whether one or two objects per hemifield are presented (*Question 2c*). In addition, following the feature-specific pattern observed in Experiment 1, we examined whether any feature-load effect in Experiment 2 was uniform across probe types or instead appeared selectively for particular probed features or load conditions. If maintaining multiple features imposes additional demands, then accuracy should be lower and CDA amplitude should be larger when participants attend to two features than when they attend to one feature within the same number-of-objects condition.

Methods

Participants

A new sample of 24 participants from the University of Missouri participated for payment. We set a target sample size comparable to Experiment 1 ($N = 20$) but adapted the sample size to accommodate counterbalancing the six experiment versions generated by the new task design. Inclusion criteria were identical to Experiment 1. Participants were replaced if their EEG data had an artifact rejection rate exceeding 25% (e.g., due to eye blinks, eye movements, and muscular activity). Five participants were replaced for this reason. After these replacements, data from 24 participants were retained for the final analysis (7 men, 17 women, $M_{age} = 25.83$ years).

Design and Procedure

Experiment 2 closely followed the design and procedures of Experiment 1, with the following key modifications.

First, the objects in the memory array were simplified. Only color and stripe orientation were used, eliminating shape to avoid potential confounding effects of stimulus complexity on working memory performance. Second, we adapted the design to use either one or two objects in the memory array. In 1-object conditions, a single stimulus appeared on each side of fixation as in Experiment 1. In 2-object conditions, two stimuli were presented on each cued side. As a result, six experimental blocks were administered: three 1-object conditions (Attend-color, Attend-stripe, Attend-both-features) and three 2-object conditions (Attend-color, Attend-stripe, Attend-both-features). (The separation of 1- and 2-object trials into different trial blocks could have allowed participants to use different encoding strategies for 1 versus 2 objects but, once the objects first are seen, different encoding strategies could be used even if blocks were mixed; with separate trial blocks, we assumed that encoding would be closer to optimal for both set sizes, allowing us to achieve the maximal object encoding.) Each stimulus subtended the same visual angle as in Experiment 1 and was centered 5° from the central fixation. When two stimuli were presented on the same side, their centers were vertically separated by 2° of visual angle to avoid overlap, identical to Wilson et al.'s parameter.

Each trial block began with 24 practice trials followed by an experimental block of 192 trials identical in timing and response requirements to those of Experiment 1. The practice blocks were shortened to 24 trials because Experiment 2 contained six critical conditions as opposed the five conditions in Experiment 1 and this kept the total session duration under approximately 3 hours. General task instructions were the same as in Experiment 1, but before each block participants were informed which feature(s) and how many objects they should remember for that block. Identical response collection, electrophysiological recording and analysis, artifact reduction precautions, and behavioral analysis were used. On average, 12.9% of trials were

rejected due to EEG artifacts such as eye movements and eye blinks. After artifact rejection and exclusion of incorrect trials, the average number of valid trials retained for CDA analysis was 152 out of 192 trials per participant in each condition.

Results

Behavioral Accuracy

Descriptive results: Mean accuracies across experimental conditions are presented in Table 2. Performance was high across 1-object conditions: Attend-color ($M = 0.95$, $SD = 0.04$), Attend-stripe ($M = 0.94$, $SD = 0.04$), Attend-both-features ($M = 0.93$, $SD = 0.05$); and relatively lower accuracy in 2-object conditions: Attend-color ($M = 0.93$, $SD = 0.04$), Attend-stripe ($M = 0.87$, $SD = 0.06$), and Attend-both-features ($M = 0.86$, $SD = 0.07$). To equate potential differences in feature difficulty, we also computed Pooled conditions that collapsed across Attend-color and Attend-stripe trials within each number of objects condition (1-pooled: $M = 0.94$, $SD = 0.03$; 2-pooled: $M = 0.90$, $SD = 0.04$).

Table 2

Mean Accuracy and Mean CDA Amplitude for Each Probed Feature in the 1-object and 2-object Conditions in Experiment 2

Condition	Probe Feature	1-object			
		Attend-1-feature		Attend-both-features	
		Accuracy	CDA	Accuracy	CDA
1-Object	Color	0.95(0.04)	-0.32(0.64)	0.94(0.05)	-0.68(0.68)
	Stripe	0.94(0.04)	-0.50(0.53)	0.92(0.06)	-0.58(0.41)
	Pooled	0.94(0.03)	-0.41(0.51)	0.93(0.05)	-0.63(0.46)
2-object	Color	0.93(0.04)	-0.75(0.52)	0.89(0.07)	-1.00(0.59)
	Stripe	0.87(0.06)	-0.97(0.70)	0.83(0.10)	-1.03(0.72)
	Pooled	0.90(0.04)	-0.86(0.48)	0.86(0.07)	-1.01(0.57)

Note. Values in parentheses represent the standard error of the mean for each condition.

Effects of number of attended feature and number of objects (*Question 2a*): To test whether attending to two features (vs. one) reduces accuracy, and whether this cost depends on the number of objects (1 vs. 2 objects), we compared the pooled Attend-1-feature and Attend-both-features conditions in a 2×2 repeated-measures ANOVA. There was decisive evidence for

a main effect of Number of Attended Features ($BF_{10} = 99.65$, $F(1, 23) = 9.64$, $p = .005$), indicating lower accuracy when participants attended to both features compared with pooled Attend-1-feature conditions. Likewise, the effect of number of objects was extremely strong ($BF_{10} > 10,000$, $F(1, 23) = 81.60$, $p < .001$), with higher accuracy for 1-object than 2-object displays. Bayesian analysis yielded only anecdotal evidence for an interaction ($BF_{10} = 1.25$), although the frequentist ANOVA suggested a significant effect ($F(1, 23) = 8.15$, $p = .009$).

Effects of number of objects within each feature attention condition (*Question 2b*):

Because number of objects itself is a central theoretical variable, we directly compared 1-object and 2-object displays within each feature condition. Preplanned Bayesian and frequentist paired-samples t tests were conducted separately comparing behavioral accuracy for 1-object versus 2-object displays for each feature attention condition, which allowed us to ask whether the number of objects matters within each feature condition, beyond the omnibus effects. For Attend-1-feature conditions, accuracy was lower for the 2-object condition than 1-object condition, with strong evidence in the Attend-stripe ($BF_{10} > 10,000$, $t(23) = 6.70$, $p < .001$) and moderate evidence in the Attend-color condition ($BF_{10} = 6.52$, $t(23) = 2.96$, $p = .007$).

For the Attend-both-features condition, trials were split by probe feature type (Color-probed vs. Stripe-probed), and each 1-object subcondition was compared with its corresponding 2-object subcondition (holding probe feature type constant). Accuracy was again lower for 2-object than 1-object displays in Color-probed trials ($BF_{10} = 7.73$, $t(23) = 3.05$, $p = .006$) and stripe-probe trials ($BF_{10} > 10,000$, $t(23) = 6.50$, $p < .001$). These findings confirm that accuracy costs of increasing the number of objects is robust across all feature conditions.

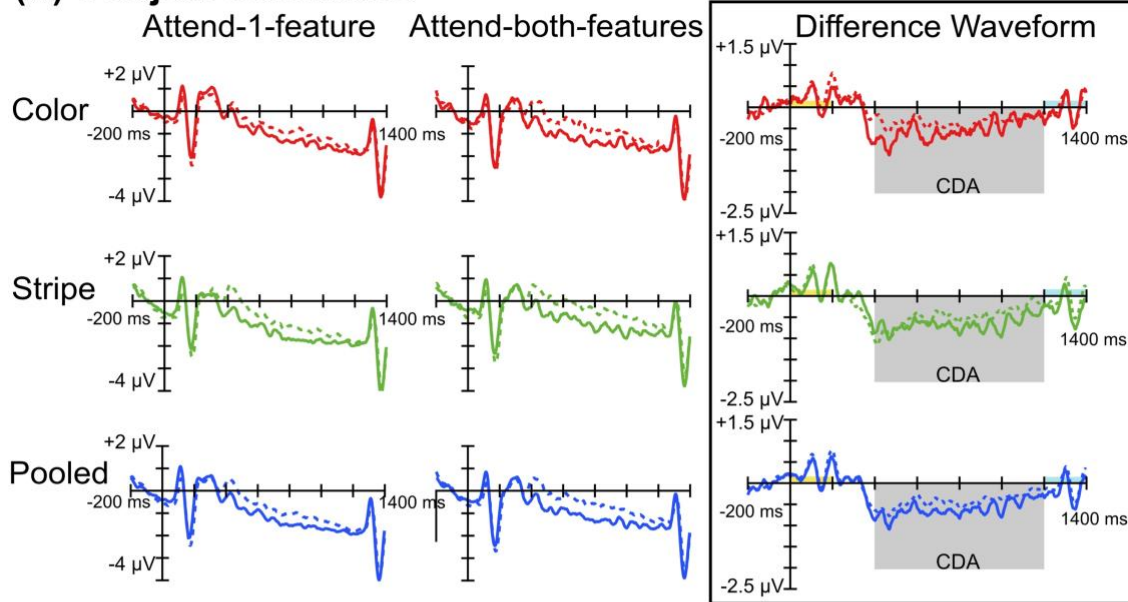
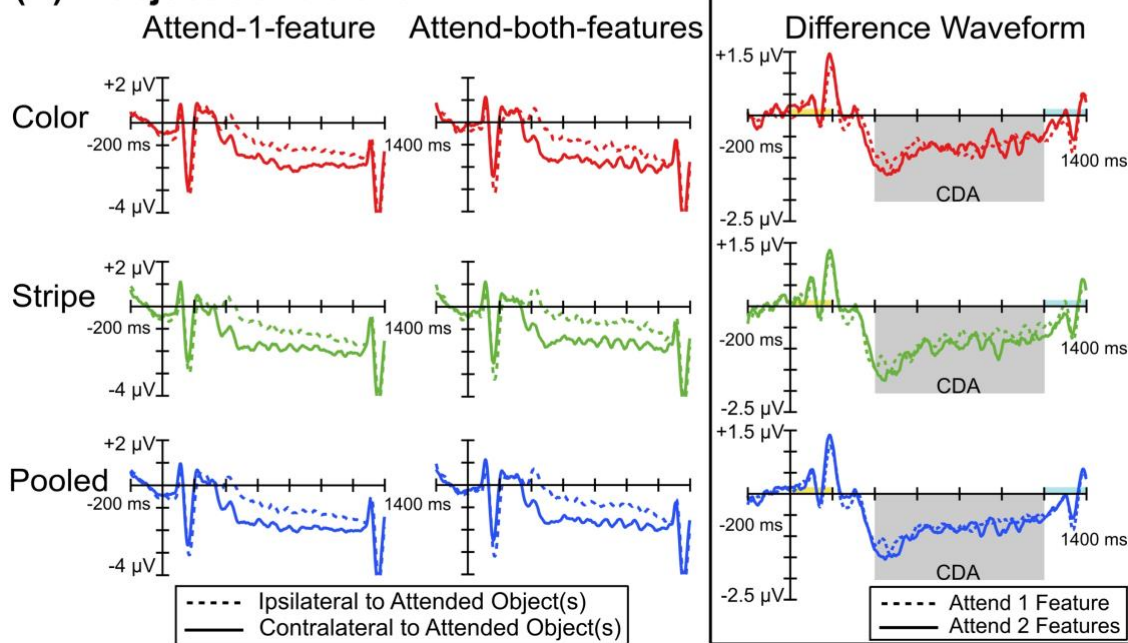
Comparisons of feature cost controlling for feature type (*Question 2c*): We next asked whether adding a second to-be-remembered feature impairs working memory performance when

the number of objects is fixed. Accuracy did not differ between Pooled Attend-1-feature and Attend-both-features conditions for 1-object trials ($BF_{10} = 0.94$, $t(23) = 1.86$, $p = .076$), but for 2-object trials, accuracy was decisively lower in the Attend-both-features condition than in the Pooled condition ($BF_{10} = 17.45$, $t(23) = 3.44$, $p = 0.002$). These results indicate that the cost of adding one more feature might be especially evident under higher object demands.

We split the Attend-both-features condition into Color-probed and Stripe-probed trials and compared each with its Attend-1-feature baseline. Within 1-object displays, Bayesian paired t tests showed no reliable feature cost: Attend-color vs. Attend-both (probe color) ($BF_{10} = 0.51$, $t(23) = 1.41$, $p = .173$) and Attend-stripe vs. Attend-both (probe stripe) ($BF_{10} = 0.48$, $t(23) = -1.36$, $p = .188$). In contrast, within 2-object displays, accuracy was significantly lower in Attend-both (probe color) compared to Attend-color ($BF_{10} = 11.57$, $t(23) = 3.24$, $p = .004$) and in Attend-both (probe stripe) compared to Attend-stripe ($BF_{10} = 3.24$, $t(23) = 2.60$, $p = .016$). These findings indicate that with larger object load, maintaining both features impairs performance relative to focusing on a single feature, regardless of whether color or stripe was probed.

CDA amplitude

Descriptive results: Figure 6 shows the grand-averaged ERP waveforms time-locked to the onset of the memory array as a function of number of objects (1-object and 2-objects) and number of features attended (Attend-1-feature or Attend-both-features). ERP waveforms are additionally separated as a function of which feature was probed (i.e., color, stripe, or pooled across both feature types). Contra-minus-difference waveforms are depicted for each condition for a comparison of the CDA component across conditions. In addition, Table 2 depicts mean CDA amplitudes for each respective condition.

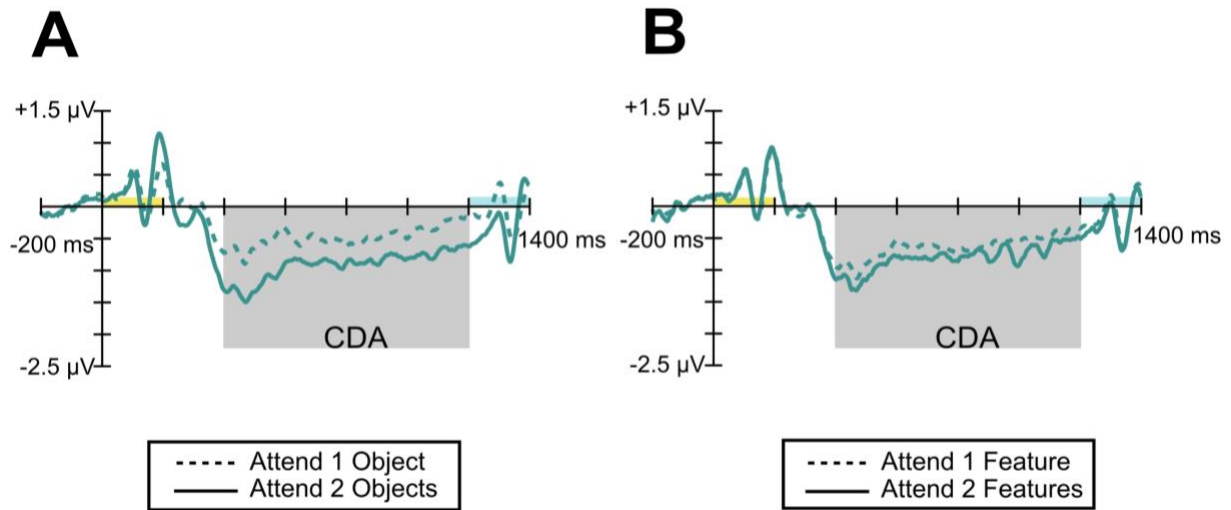
Figure 6*ERP Waveforms for Each Condition Separately and Pooled Across Features in Experiment 2***(A) 1-object Conditions****(B) 2-object Conditions**

Note. ERP waveforms are shown separately for 1-object conditions (Panel A) and 2-object conditions (Panel B). For the two probe feature types (Color, Stripe) separately and across both types pooled, the left and middle columns show grand averaged ERP waveforms recorded at posterior electrode sites (PO7/PO8), plotted separately for Attend-1-feature and Attend-both-features conditions. Solid lines represent activity contralateral to the attended object, and dashed lines represent ipsilateral activity. The right column shows the corresponding difference waveforms, with the CDA time window highlighted in gray. The yellow bar on the X axis indicates the memory array presentation period (0-200 ms), and the blue bar on the X axis at the end of the epoch marks the onset of the response period (1200-1400 ms). All waveforms were low-pass filtered at 20 Hz for visualization purposes only; all statistical analyses were conducted without the low-pass filter.

As shown, CDA waveforms showed little variation as a function of the number of attended features. Across feature types, the Attend-1-feature and Attend-both-features conditions produced highly similar CDA patterns. Below, we will examine the CDA amplitude using a series of analyses that parallel those conducted for behavioral performance above.

Effects of number of attended feature and number of objects (*Question 2a*): The 2×2 ANOVA on mean CDA amplitude revealed no reliable interaction between Number of Attended features (Pooled vs. Attend-both) and Number of Objects ($BF_{10} = 0.32$, $F(1, 23) = 0.40$, $p = .532$). However, there was decisive evidence for a main effect of Number of Objects ($BF_{10} > 10,000$, $F(1, 23) = 21.93$, $p < .001$), reflecting larger CDA amplitudes for two than one object overall. Bayesian results provided modest evidence for a main effect of Number of Attended features ($BF_{10} = 2.02$, $F(1, 23) = 4.45$, $p = .046$), suggesting a relatively small effect of feature load.

To visualize this effect, Figure 7 plots CDA difference waveforms by the number of objects, collapsing across the number of attended features (Figure 7A), and by the number of attended features, collapsing across the number of attended objects (Figure 7B). As shown, the CDA is clearly larger when participants attended two objects than when they attended one object. In comparison, the CDA is only slightly larger when participants attended two features than when they attended one feature. This visualization highlights that the number of objects has a stronger influence on CDA amplitude than the number of attended features, consistent with object-based accounts (e.g., Luria & Vogel, 2011; Quak et al., 2018). Although we cannot entirely rule out a feature-load effect on the CDA (Wilson et al., 2012), this comparison does show that feature-load effects were small in comparison to object-load effects (see also next sections).

Figure 7*CDA Waveforms as a Function of Object Load and Feature Load in Experiment 2*

Note. Panel A shows CDA difference waveforms as a function of object load (i.e., for attending 1 object versus 2 objects, collapsing across the number of attended features). Panel B shows CDA difference waveforms as a function of feature load (i.e., for attending 1 feature versus 2 features, collapsing across the number of attended objects). The CDA time window is shaded in gray. The yellow bar on the X axis indicates the memory array period from 0 to 200 ms, and the blue bar on the X axis indicates the onset of response period from 1200 to 1400 ms.

Effects of number of objects within each feature condition (*Question 2b*): Pairwise

comparisons were applied to examine the effect of number of objects (1 vs. 2) within each feature attention condition. In Attend-1-feature conditions, CDA amplitude was greater for two than for one object in the Attend-color condition ($BF_{10} = 8.98$, $t(23) = 3.12$, $p = .005$) and Attend-stripe condition ($BF_{10} = 16.36$, $t(23) = 3.41$, $p = .002$).

For the Attend-both-features condition, trials were split by probe feature type (Color-probed vs. Stripe-probed). Each 1-object subcondition was then compared with its corresponding 2-object subcondition, holding probe feature type constant. In Stripe-probed trials, CDA amplitude was again larger for two than for one object ($BF_{10} = 11.79$, $t(23) = 3.25$, $p = .004$). In contrast, in Color-probed trials, Bayesian evidence was weak ($BF_{10} = 1.24$) and the frequentist test was marginal ($t(23) = 2.04$, $p = .05$), providing only limited support for an object effect in

this subcondition. Across most feature conditions, CDA amplitude was consistently higher in 2-object than 1-object conditions, indicating robust sensitivity to the number of maintained objects.

Comparisons of feature cost controlling for feature type (*Question 2c*): We asked whether adding a second to-be-attended feature modulates CDA amplitude when the number of objects is held constant. Bayesian t tests indicated no compelling evidence for differences between Pooled and Attend-both-features conditions within either 1-object ($BF_{10} = 1.00$, $t(23) = 1.90$, $p = .070$) or 2-object ($BF_{10} = 0.65$, $t(23) = 1.59$, $p = .125$) displays. Thus, unlike behavioral accuracy, CDA amplitude appeared primarily sensitive to number of objects, with little to no reliable evidence for additional modulation by feature load.

We divided the Attend-both-features condition into Color-probed and Stripe-probed trials and compared each with its corresponding Attend-1-feature baseline, holding both probe feature type and number of objects constant. For Color-probe trials, there was only modest evidence for a difference between Attend-color and Attend-both-features conditions within 1-object ($BF_{10} = 2.22$, $t(23) = 2.39$, $p = .026$), but weak evidence favoring no difference within 2-object ($BF_{10} = 0.73$, $t(23) = 1.68$, $p = .106$) conditions. For Stripe-probe trials, Bayes factors favored the null for comparisons between Attend-color and Attend-both-features conditions within 1-object ($BF_{10} = 0.27$, $t(23) = 0.74$, $p = .468$) and 2-object ($BF_{10} = 0.24$, $t(23) = 0.46$, $p = .648$) conditions.

Overall, these results provide little consistent evidence that CDA amplitude increases with feature load once probe feature type and number of objects are controlled. Any feature-load effect appears limited and weak, emerging only for 1-object color-probed trials (where the evidence was modest). The absence of CDA differences between Attend-1-feature (Attend-color and Attend-stripe) versus Attend-both-features conditions here likely depends on the fact that here the conditions were matched for perceptual complexity.

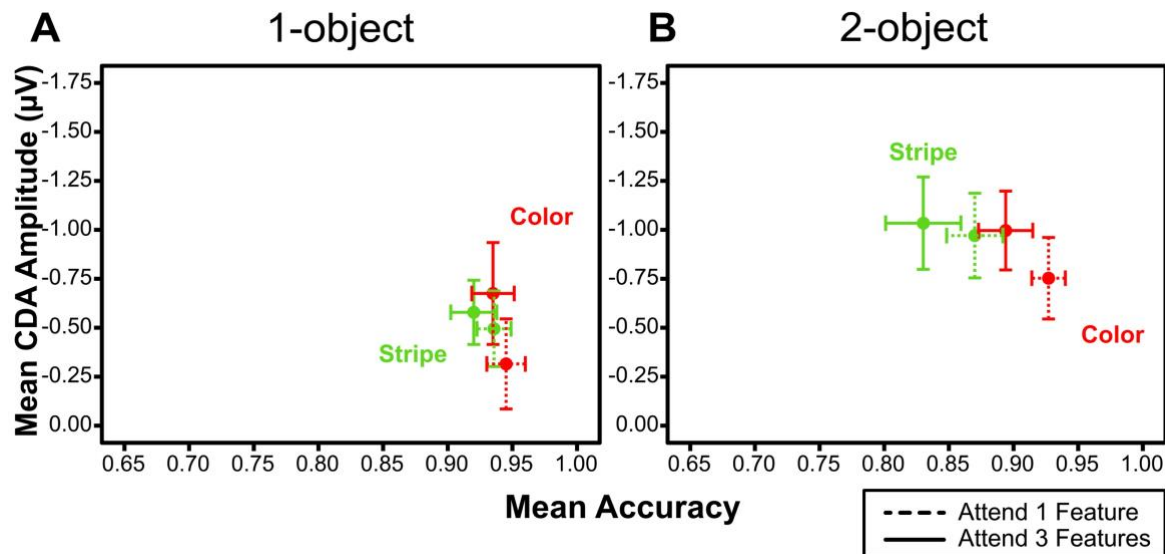
Relationship between Behavioral Accuracy and CDA Amplitude

We plotted mean CDA amplitude against mean accuracy for each condition (Figure 8). Rather than forming a single monotonic relationship, the data revealed distinct clustering patterns as a function of number of objects.

In the 1-object conditions (Figure 8A), data points clustered tightly at high accuracy values, with relatively small CDA amplitudes. Within this cluster, variation across feature conditions was modest, except that attention to color alone resulted in a smaller CDA than attention to all features, despite little behavioral difference (replicating the pattern obtained in Experiment 1).

Figure 8

Relation Between Behavioral Accuracy and CDA Amplitude Across Conditions in Experiment 2



Note. Each point is a condition mean. Error bars denote within-subject 95% CIs (Morey, 2008).

The 2-object conditions (Figure 8B) formed a clearly separate cluster characterized by more negative CDA amplitudes and overall lower accuracy than the 1-object conditions. Within this cluster, feature conditions showed some dispersion along the accuracy dimension, but CDA amplitudes remained comparably large, suggesting that increases in neural maintenance activity

were primarily associated with the number of objects rather than with feature-specific performance differences.

Negative Slow Waveform (NSW)

As in Experiment 1, we conducted an exploratory analysis of the NSW as an additional measure of working memory in the online Supplementary Materials (Fukuda et al., 2015; Feldman-Wüstefeld, 2021). We examined whether NSW amplitude was modulated by object load (1-object vs. 2-objects), feature load (Attend-1-feature vs. Attend-2-features), or their interaction. The analysis revealed a clear object-load effect on the NSW, $BF_{10} = 69.40$; $t(19) = 8.84$; $p = .007$, but not feature-load effect on the NSW, $BF_{10} = 0.37$; $t(19) = 1.36$; $p = .255$, or interaction between feature load and object load, $BF_{10} = 0.38$; $t(19) = 1.23$; $p = .28$. Thus, consistent with the CDA results, the NSW was sensitive to object load but not feature load.

Exploratory Decoding Analysis

Two exploratory EEG decoding analyses complemented the CDA and behavioral findings (Adam et al., 2020; Jones et al., 2024, see Supplementary Materials). A classifier trained to distinguish 1-object, pooled Attend-1-feature condition from 2-object, pooled Attend-1-feature condition placed the untrained 1-object, Attend-both-feature condition on the one-object side of the decision boundary, consistent with an object-based load signal. Separate classifiers decoded both feature load and object load during retention, with no reliable difference in mean accuracy. Feature-load decoding may, however, partly reflect differences in attention or effort rather than feature storage alone.

Discussion

Experiment 2 was designed to further disentangle the relative contributions of number of objects and number of features in visual working memory. Here, several key findings emerged.

First, behavioral accuracy clearly demonstrated both effects of number of objects and number of features. Accuracy was reliably reduced when two objects had to be stored compared to one, and this effect was magnified when participants were required to retain both color and stripe rather than a single feature. Importantly, splitting the Attend-both-features condition by probe feature types revealed that these costs of feature load on accuracy were especially robust under two-object conditions. Specifically, both color- and stripe-probed trials showed reduced accuracy when two features were attended, whereas no reliable feature costs were present when only one object was maintained (i.e., in the one-object condition). Altogether, the behavioral results indicate that the effect of feature load primarily manifests when memory demands are compounded by the need to distribute attention across multiple objects.

Second, the CDA results presented a contrasting pattern, in that CDA amplitude was generally more sensitive to number of objects than to number of features. This pattern is broadly consistent with prior studies linking CDA amplitude to the number of integrated objects maintained in memory (Luria & Vogel, 2011; Woodman & Vogel, 2008). At the same time, the probe-specific analyses suggested that this object-dominant pattern was not equally strong across all subconditions. The effect of number of objects was clear for stripe-probed trials, but only weakly supported for color-probed trials. Likewise, evidence for a feature-load effect on CDA amplitude was limited and appeared only in color-probe trials under the 1-object condition, disappearing under 2-object conditions. Thus, Experiment 2 did not reveal a general sensitivity of CDA amplitude to feature load, but instead suggested that any such sensitivity was selective and context-dependent.

In summary, our Experiment 2 provides converging evidence that working memory accuracy is constrained by both number of objects and number of features, with effect of number

of features emerging primarily under high object-load conditions. CDA amplitude, however, was more closely related to number of objects than to number of features, while showing only limited evidence of feature-load sensitivity in color-probe trials under low object load.

General Discussion

The present study investigated the extent to which working memory capacity is constrained by the number of objects versus the number of features stored, using the CDA as a neural marker in conjunction with behavioral accuracy. Across two experiments, we manipulated the number of objects present and feature load (sometimes in terms of the number of features present and sometimes in terms of the number of features to be attended), with attention to all objects on the cued side of the display. We were able to manipulate the features to be attended while holding perceptual input constant, thereby building on and extending prior work (Wilson et al., 2012). Overall, our findings were broadly consistent with prior object-based accounts of the CDA (e.g., Quak et al., 2018), but they also qualified that conclusion in an important way.

Across both experiments, behavioral accuracy reflected costs of both object load and feature load, as in previous work (e.g., Hardman & Cowan, 2015; Oberauer & Eichenberger, 2013). In contrast, the CDA amplitude was more closely tied to object load overall. At the same time, probe-specific analyses suggested that this object-dominant CDA pattern was not completely uniform, because limited feature-load effects emerged when attention was to be directed to color versus to color plus other features, specifically in the case in which there was one object to be attended.

Our results are not totally in keeping with any of the previous findings. First, whereas Wilson et al.'s (2012) proposed that CDA reflect both object load and feature load for the PO7/PO8 electrode pair, our results showed a strong effect of object load and only limited,

feature-specific evidence of feature-load modulation. Second, the findings are not totally in keeping with the object-based account of Quak et al. (2018). Our results suggest that if Quak et al. had separately examined attention to color versus all four features they used, they would have seen a feature load effect. Third, although our results are most similar to those of Woodman and Vogel (2008), there are still differences between the results of the studies. In particular, Woodman and Vogel and the present study both found that attention to color yields a smaller CDA than attention to multiple features. However, Woodman and Vogel found this effect across different levels of object load, whereas in our study it emerged (in both experiments) for one attended object, but did not emerge (in Experiment 2) for two attended objects.

Our Experiment 2 resembles the study of Woodman and Vogel (2008) in that we manipulated both the number of objects and number of features to be attended and remembered. Like our study, Woodman and Vogel concluded that the CDA was dominated by the object load. However, their results also suggested a more specific type of feature selectivity that should be distinguished from the broader feature load effect reported by Wilson et al. (2012). Specifically, attention to color alone yielded a smaller CDA than attention to color plus orientation or, in another experiment, attention to color plus shape. Attention to either of these other features alone, however, yielded a CDA comparable to attention to color plus the other feature. Thus, Woodman and Vogel's pattern is best understood not as evidence that feature load adds up across all attended features, but as evidence that selective attention to one feature dimension can reduce CDA when that feature is easier than the others (e.g., if that feature is processed more automatically than others).

Importantly, the present results suggest that a similar but limited selective pattern can emerge under low object load. In our 1-object condition, attending to color alone yielded a

smaller CDA than attending to color plus orientation and shape in Experiment 1, or color plus orientation in Experiment 2. However, unlike Woodman and Vogel (2008), we did not observe this selective effect under higher object load. Whereas they found it with 2-object and 4-object displays, we did not find it in our 2-object condition. Thus, the selective color effect was restricted to a one-object-load condition in our study.

One possible explanation for this discrepancy between our results and those of Woodman and Vogel (2008) lies in differences in color feature values. Woodman and Vogel used seven colors but only four orientations, whereas we used four values for each feature dimension. Larger and more diverse feature sets are known to reduce trial-to-trial proactive interference, whereas smaller and more repetitive feature sets can produce stronger interference across trials. Stronger interference may differentially affect working memory performance (Mercer et al., 2022). In addition, Woodman and Vogel selected feature values without replacement on a trial, whereas our study sampled with replacement. Woodman and Vogel's use of seven colors and selection of colors without replacement on a trial could work together to make the Attend-color trials more distinct from one another than in our study, potentially reducing the CDA amplitude for attending colors.

Our results show that when the difference between color and other features is not as large as was the case in Woodman and Vogel (2008), the difference between attention to color alone versus multiple features becomes more difficult to detect. We suggest that it can be detected in the one-object condition of our study perhaps because the CDA is then far from its asymptotic level, leaving more room in the response for effects based on feature attention to be observed (see also Bays & Husain, 2008). Under higher object-load conditions, CDA amplitude may be driven more strongly by the number of maintained object files, reducing the visibility of feature-

specific variation. It is alternatively possible that, as load increases, representations are maintained in a more integrated object-level format, which would further reduce the detectability of feature-specific differences in the CDA.

Under the present task conditions, color was the feature dimension for which the distinction between single-feature and multi-feature maintenance was most readily expressed. These findings suggest an *object-load feature-difficulty hypothesis*, according to which the CDA is primarily determined by the number of maintained objects, while feature-related effects emerge selectively when an easier single feature can be contrasted with attention to multiple features, of which at least one is more difficult. We suggest that the reason the effect of feature attention has been limited to the color feature is that that feature is easier to process than the other features. Based on Woodman and Vogel (2008) and the present study, we surmise that the CDA does not sum across attended features but rises to the level of the most difficult feature that needs to be processed, which will form a contrast to the lower CDA of attention to color alone. (It is noteworthy that Wilson et al., 2012 also pointed out the apparent ease of colors relative to orientation and shape features). It is important to highlight that this is only a hypothesis, and the present study did not directly manipulate feature difficulty. It is also plausible that certain feature dimensions might have been more verbalizable or separable from others which could have modulated CDA amplitude. It will therefore be important for future research to determine whether other features can be manipulated to make them easier to process, in which case these other features could show the same feature-based attention effect.

Another finding warranting comment is that of Gao et al. (2011). They required memory of the orientations of 2 or 4 arrows on a trial, defined as low or high resolution depending on whether each recognition lure in a trial block was very different from the orientation of the item

presented at the location of the lure (requiring only low resolution) or just slightly different from the presented item (requiring high resolution). The CDA showed an object effect only in the low-resolution condition; the CDA for 4 low-resolution objects was higher than the CDA for the other conditions (2 low-resolution and 2 or 4 high-resolution). The result shows that the need to extract fine-grained information from each item played an important role and not just the number of objects to be remembered. In this case, when high-resolution information was needed, participants may not have tried to retain all 4 items. Conversely, in our Experiment 2, when 2 multifeatured objects were to be remembered, participants may not have tried to remember all features and thus there was no attend-color-only decrement in CDA with 2 attended objects. An important topic for future research to reconcile these studies is whether attention can be allocated either to some details of remembered complex objects at the expense of forgetting other objects entirely (perhaps the case in Gao et al., 2011) or forgetting some features to retain more than one complex object (perhaps the case in our Experiment 2).

Finally, the plots directly comparing behavioral accuracy and CDA amplitude across conditions (Figure 4 and 7) highlight a dissociation between the two factors. Behavioral accuracy was sensitive to both object load and feature load, whereas CDA amplitude mainly tracked object load and showed only limited evidence of feature-specific modulation. This dissociation is consistent with Quak et al. (2018) and suggests that behavioral measures and CDA amplitude do not index identical aspects of working memory. Behavioral performance likely reflects multiple sources of limitation, including feature encoding and decision demands, whereas the CDA more selectively indexes the maintenance of individuated objects. As the immediately following section suggests, the difference between measures could be that the CDA does not reflect the

portion of processing that distinguishes in fine detail between all of the task-relevant versus task-irrelevant features of objects in working memory.

Incidental Encoding of Irrelevant Features

One potential limitation that should be acknowledged is the possibility of incidental encoding of irrelevant features. In the present study, the CDA was largely insensitive to the number of attended features. A potential explanation for this finding is that all features on an object may have been automatically encoded, regardless of task relevance (see Marshall & Bays, 2013). Although there is evidence that irrelevant objects can be suppressed (Feldmann-Wüstefeld & Vogel, 2019; Gaspelin et al., 2025) and that the CDA reflects working memory load, not visual attention per se (Feldmann-Wüstefeld et al., 2018), these findings do not necessarily apply to features to be ignored within objects to be attended. Incidental encoding of irrelevant features into working memory could explain why our results differed from Wilson et al. (2012): in their study, the Attend-1-feature conditions presented single-featured objects in isolation, which would have prevented incidental encoding of task-irrelevant features. Thus, although the present study directly addresses the issue of a perceptual confound in Wilson et al.'s (2012) design by using multifeatured objects and requiring attention to one or all features, it is also possible that the revised approach opens the possibility of incidental feature encoding. Indeed, or objects with only stripe orientation produced smaller CDAs compared to attention to stripe orientation alone in multifeatured objects.

Some of our findings, however, are inconsistent with an incidental encoding account. First, the behavioral data showed clear effects of the number of attended features, suggesting that participants were able to encode certain features selectively, at least to some degree. If participants mandatorily encoded all irrelevant features, behavioral memory performance should

have been the same irrespective of the number of features attended. Second, and as explained in previous sections, attention to certain features (such as color) did modulate the CDA amplitude, and this type of feature-based modulation should not occur if all features must be mandatorily encoded.

In all, the data point to a situation in which the CDA reflects incidental encoding of features to some degree, along with further enhanced encoding of attended features as in Woodman & Vogel (2008). It may be, moreover, that the attended-feature extra encoding enhances the CDA only for some features or combination of features. For example, it may be that shape and orientation are perceptually integral features that make either of them difficult to extract from an object, whereas color may be perceptually separable from these features and easier to extract (Cheng & Pachella, 1984; Garner, 1976). That is a hypothesis that remains to be assessed in future work. Other research hypotheses regarding incidental encoding of features also should be examined (e.g., whether attention to color is still possible for shades within a conventional color category and, conversely, whether attention to orientations might become possible if the orientations are made more explicitly categorical, as has been the case with colors).

In any case, future studies are needed to address the possibility of incidental encoding. In our view, simply presenting lone features (i.e., without task-irrelevant features) would not suffice because this would allow perceptual level effects to contaminate the ERPs and therefore make it difficult to interpret differences between conditions attributable to working memory maintenance (for evidence of this, see the solo-stripe condition of Figure 3). Instead, we advocate a more systematic investigation of factors affecting the efficacy of feature-based attention in both behavioral and CDA measures.

Conclusion and Future Directions

The present study provides new evidence regarding the mechanisms that constrain working memory. Across two experiments, behavioral results consistently showed reliable costs of both number of objects and number of features. The effects of number of objects were robust across conditions, whereas CDA effects of feature load appeared only in limited and selective contexts. Specifically, attending to one color produced a smaller CDA than attending to multiple features including color, but only in the context of a single attended object.

The present findings therefore make two main contributions. First, they strengthen the conclusion from prior CDA research that object-based storage is the dominant pattern in this neural measure. Second, they refine that conclusion by showing that feature-load effects on the CDA, when they occur, may be selective and context-dependent rather than broad and uniform. The present findings help situate the current study within an existing spectrum of CDA results, one that ranges from more strongly feature-based to more strongly object-based outcomes. In that context, our results using more strict design appear closer to the intermediate position represented by Woodman and Vogel (2008) than to the more purely object-based end represented by Quak et al. (2018) or feature-based CDA effect suggested by Wilson et al. (2012).

Future work should examine these boundary conditions more directly, including whether feature-specific CDA effects depend on object load, feature type, spatial configuration, or the discriminability of particular feature dimensions. It will also be useful to combine the CDA with other neural measures to test whether feature information is represented neurally even when it does not strongly modulate CDA amplitude.

References

- Adam, K. C. S., Vogel, E. K., & Awh, E. (2020). Multivariate analysis reveals a generalizable human electrophysiological signature of working memory load. *Psychophysiology*, 57(12), e13691. <https://doi.org/10.1111/psyp.13691>
- Alvarez, G. A., & Cavanagh, P. (2004). The capacity of visual short-term memory is set both by visual information load and by number of objects. *Psychological science*, 15(2), 106-111. <https://doi.org/10.1111/j.0963-7214.2004.01502006.x>
- Barense, M. D., Bussey, T. J., Lee, A. C., Rogers, T. T., Davies, R. R., Saksida, L. M., ... & Graham, K. S. (2005). Functional specialization in the human medial temporal lobe. *Journal of Neuroscience*, 25(44), 10239-10246. <https://doi.org/10.1523/JNEUROSCI.2704-05.2005>
- Bays, P. M., & Husain, M. (2008). Dynamic shifts of limited working memory resources in human vision. *Science*, 321(5890), 851-854. <https://doi.org/10.1126/science.1158023>
- Berggren, N., & Eimer, M. (2016). Does contralateral delay activity reflect working memory storage or the current focus of spatial attention within visual working memory?. *Journal of Cognitive Neuroscience*, 28(12), 2003-2020. https://doi.org/10.1162/jocn_a_01019
- Cai, Y., Fulvio, J. M., Samaha, J., & Postle, B. R. (2022). Context binding in visual working memory is reflected in bilateral event-related potentials, but not in contralateral delay activity. *eneuro*, 9(6), ENEURO-0207. <https://doi.org/10.1523/ENEURO.0207-22.2022>
- Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24(1), 87-114. <https://doi.org/10.1017/S0140525X01003922>

- Cowan, N. (1988). Evolving conceptions of memory storage, selective attention, and their mutual constraints within the human information processing system. *Psychological Bulletin*, 104, 163-191. <https://doi.org/10.1037/0033-2909.104.2.163>
- Cowan, N., Bao, C., Bishop-Chrzanowski, B. M., Costa, A. N., Greene, N. R., Guitard, D., ... & Ünal, Z. E. (2024). The relation between attention and memory. *Annual Review of Psychology*, 75(1), 183-214. <https://doi.org/10.1146/annurev-psych-040723-012736>
- Cowan, N., Blume, C. L., & Saults, J. S. (2013). Attention to attributes and objects in working memory. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 39, 731-747. <https://doi.org/10.1037/a0029687>
- Cheng, P. W., & Pachella, R. G. (1984). A psychophysical approach to dimensional separability. *Cognitive Psychology*, 16(3), 279–304. [https://doi.org/10.1016/0010-0285\(84\)90011-2](https://doi.org/10.1016/0010-0285(84)90011-2)
- Delorme, A., & Makeig, S. (2004). EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *Journal of neuroscience methods*, 134(1), 9-21. <https://doi.org/10.1016/j.jneumeth.2003.10.009>
- Feldmann-Wüstefeld, T. (2021). Neural measures of working memory in a bilateral change detection task. *Psychophysiology*, 58(1), e13683. <https://doi.org/10.1111/psyp.13683>
- Feldmann-Wüstefeld, T., & Vogel, E. K. (2019). Neural evidence for the contribution of active suppression during working memory filtering. *Cerebral Cortex*, 29(2), 529-543. <https://doi.org/10.1093/cercor/bhx336>

- Feldmann-Wüstefeld, T., Vogel, E. K., & Awh, E. (2018). Contralateral delay activity indexes working memory storage, not the current focus of spatial attention. *Journal of cognitive neuroscience*, 30(8), 1185-1196. https://doi.org/10.1162/jocn_a_01271
- Fukuda, K., Mance, I., & Vogel, E. K. (2015). α power modulation and event-related slow wave provide dissociable correlates of visual working memory. *Journal of Neuroscience*, 35(41), 14009-14016. <https://doi.org/10.1523/JNEUROSCI.5003-14.2015>
- Forsberg, A., Adams, E. J., & Cowan, N. (2023). Why does visual working memory ability improve with age: More objects, more feature detail, or both? A registered report. *Developmental science*, 26(2), e13283. <https://doi.org/10.1111/desc.13283>
- Gao, Z., Li, J., Liang, J., Chen, H., Yin, J., & Shen, M. (2009). Storing fine detailed information in visual working memory—evidence from event-related potentials. *Journal of Vision*, 9(7), 17, 1–12. <https://doi.org/10.1167/9.7.17>
- Gao, Z., Yin, J., Xu, H., Shui, R., & Shen, M. (2011). Tracking object number or information load in visual working memory: revisiting the cognitive implication of contralateral delay activity. *Biological Psychology*, 87(2), 296-302. <https://doi.org/10.1016/j.biopsycho.2011.03.013>
- Garner, W. R. (1976). Interaction of stimulus dimensions in concept and choice processes. *Cognitive Psychology*, 8(1), 98-123. [https://doi.org/10.1016/0010-0285\(76\)90006-2](https://doi.org/10.1016/0010-0285(76)90006-2)
- Gaspar, J. M., Christie, G. J., Prime, D. J., Jolicœur, P., & McDonald, J. J. (2016). Inability to suppress salient distractors predicts low visual working memory capacity, *PNAS*, 113 (13) 3693-3698, <https://doi.org/10.1073/pnas.1523471113> (2016).

- Gaspelin, N., & Cowan, N. (2025). Restoring a top-down control assumption: Salience effects in working memory are overcome with time. *Journal of Experimental Psychology: General*, 154(8), 2301–2317. <https://doi.org/10.1037/xge0001776>
- Gaspelin, N., & Luck, S. J. (2018). Electrophysiological and behavioral evidence of suppression of salient-but-irrelevant stimuli. *Journal of Cognitive Neuroscience*, 30(9), 1265–1280. https://doi.org/10.1162/jocn_a_01279
- Gaspelin, N., Ma, X., & Luck, S. J. (2025). Signal suppression 2.0: An updated account of attentional capture and suppression. *Psychonomic Bulletin & Review*, 32(6), 2648-2668. <https://doi.org/10.1111/1467-9280.01458>
- Hardman, K. O., & Cowan, N. (2015). Remembering complex objects in visual working memory: Do capacity limits restrict objects or features?. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 41(2), 325. <http://dx.doi.org/10.1037/xlm0000031>
- Jones, H. M., Diaz, G. K., Ngiam, W. X. Q., & Awh, E. (2024). Electroencephalogram decoding reveals distinct processes for directing spatial attention and encoding into working memory. *Psychological Science*, 35(10), 1108-1138. <https://doi.org/10.1177/09567976241263002>
- Kleiner, M., Brainard, D., & Pelli, D. (2007). *What's new in Psychtoolbox-3?*.
- Lins, O. G., Picton, T. W., Berg, P., & Scherg, M. (1993). Ocular artifacts in EEG and event-related potentials I: Scalp topography. *Brain topography*, 6, 51-63. <https://doi.org/10.1007/BF01234127>

- Lopez-Calderon, J., & Luck, S. J. (2014). ERPLAB: an open-source toolbox for the analysis of event-related potentials. *Frontiers in human neuroscience*, 8, 213.
<https://doi.org/10.3389/fnhum.2014.00213>
- Luck, S. J. (2014). *An introduction to the event-related potential technique*. MIT press.
- Luck, S. J., & Gaspelin, N. (2017). How to get statistically significant effects in any ERP experiment (and why you shouldn't). *Psychophysiology*, 54(1), 146–157.
<https://doi.org/10.1111/psyp.12639>
- Luck, S. J., & Vogel, E. K. (1997). The capacity of visual working memory for features and conjunctions. *Nature*, 390(6657), 279–281. <https://doi.org/10.1038/36846>
- Lüdtke, D. (2019). Package 'sjstats'. *Statistical Functions for Regression Models*, version, 17(3). Retrieved from <https://cran.r-hub.io/web/packages/sjstats/sjstats.pdf>
- Luria, R., Balaban, H., Awh, E., & Vogel, E. K. (2016). The contralateral delay activity as a neural measure of visual working memory. *Neuroscience & Biobehavioral Reviews*, 62, 100–108. <https://doi.org/10.1016/j.neubiorev.2016.01.003>
- Luria, R., Sessa, P., Gotler, A., Jolicoeur, P., & Dell'Acqua, R. (2010). Visual short-term memory capacity for simple and complex objects. *Journal of Cognitive Neuroscience*, 22(3), 496–512. <https://doi.org/10.1162/jocn.2009.21214>
- Luria, R., & Vogel, E. K. (2011). Shape and color conjunction stimuli are represented as bound objects in visual working memory. *Neuropsychologia*, 49(6), 1632–1639.
<https://doi.org/10.1016/j.neuropsychologia.2010.11.031>
- Ly, A., Raj, A., Etz, A., Marsman, M., Gronau, Q. F., & Wagenmakers, E. J. (2018). Bayesian

- reanalyses from summary statistics: a guide for academic consumers. *Advances in Methods and Practices in Psychological Science*, 1(3), 367-374.
<https://doi.org/10.1177/2515245918779348>
- Marshall, L., & Bays, P. M. (2013). Obligatory encoding of task-irrelevant features depletes working memory resources. *Journal of vision*, 13(2), 21-21.
<https://doi.org/10.1167/13.2.21>
- Morey, R. D. (2008). Confidence intervals from normalized data: A correction to Cousineau (2005). *Tutorials in quantitative methods for psychology*, 4(2), 61-64.
<https://doi.org/10.20982/tqmp.04.2.p061>
- Morey, R. D., & Rouder, J. N. (2013). Bayesfactor: Computation of bayes factors for common designs [Computer software manual]. Retrieved from <http://CRAN.R-project.org/package=BayesFactor> (R package version 0.9.5)
- Ngiam, W. X., Adam, K. C., Quirk, C., Vogel, E. K., & Awh, E. (2021). Estimating the statistical power to detect set-size effects in contralateral delay activity. *Psychophysiology*, 58(5), e13791. <https://doi.org/10.1111/psyp.13791>
- Oberauer, K., & Eichenberger, S. (2013). Visual working memory declines when more features must be remembered for each object. *Memory & cognition*, 41, 1212-1227.
<https://doi.org/10.3758/s13421-013-0333-6>
- Quak, M., Langford, Z. D., London, R. E., & Talsma, D. (2018). Contralateral delay activity does not reflect behavioral feature load in visual working memory. *Biological Psychology*, 137, 107-115. <https://doi.org/10.1016/j.biopsycho.2018.07.006>

- Rouder, J. N., Morey, R. D., Morey, C. C., & Cowan, N. (2011). How to measure working memory capacity in the change detection paradigm. *Psychonomic Bulletin & Review*, 18(2), 324–330. <https://doi.org/10.3758/s13423-011-0055-3>
- Rouder, J. N., Morey, R. D., Speckman, P. L., & Province, J. M. (2012). Default Bayes factors for ANOVA designs. *Journal of Mathematical Psychology*, 56(5), 356-374. <https://doi.org/10.1016/j.jmp.2012.08.001>
- Roy, Y., & Faubert, J. (2023). Is the Contralateral Delay Activity (CDA) a robust neural correlate for Visual Working Memory (VWM) tasks? A reproducibility study. *Psychophysiology*, 60(2), e14180. <https://doi.org/10.1111/psyp.14180>
- Stilwell, B. T., Egeth, H., & Gaspelin, N. (2022). Electrophysiological evidence for the suppression of highly salient distractors. *Journal of Cognitive Neuroscience*, 34(5), 787–805. https://doi.org/10.1162/jocn_a_01827
- Talcott, T. N., & Gaspelin, N. (2021). Eye movements are not mandatorily preceded by the N2pc component. *Psychophysiology*, e13821. <https://doi.org/10.1111/psyp.13821>
- Thaler, L., Schütz, A. C., Goodale, M. A., & Gegenfurtner, K. R. (2013). What is the best fixation target? The effect of target shape on stability of fixational eye movements. *Vision research*, 76, 31-42. <https://doi.org/10.1016/j.visres.2012.10.012>
- Vandenbroucke, A. R., Sligte, I. G., & Lamme, V. A. (2011). Manipulations of attention dissociate fragile visual short-term memory from visual working memory. *Neuropsychologia*, 49(6), 1559-1568. <https://doi.org/10.1016/j.neuropsychologia.2010.12.044>.

- Vogel, E. K., & Machizawa, M. G. (2004). Neural activity predicts individual differences in visual working memory capacity. *Nature*, 428(6984), 748-751.
<https://doi.org/10.1038/nature02447>
- Wilson, K. E., Adamo, M., Barense, M. D., & Ferber, S. (2012). To bind or not to bind: Addressing the question of object representation in visual short-term memory. *Journal of Vision*, 12(8), 14-14. <https://doi.org/10.1167/12.8.14>
- Woodman, G. F., & Luck, S. J. (2003). Serial deployment of attention during visual search. *Journal of Experimental Psychology. Human Perception and Performance*, 29(1), 121–138. <https://doi.org/10.1167/1.3.103>
- Woodman, G. F., & Vogel, E. K. (2008). Selective storage and maintenance of an object's features in visual working memory. *Psychonomic bulletin & review*, 15(1), 223-229.
<https://doi.org/10.3758/PBR.15.1.223>

**Feature-Based and Object-Based Limits in Visual Working Memory: the Nature of
Contralateral Delay Activity**

Supplementary Materials

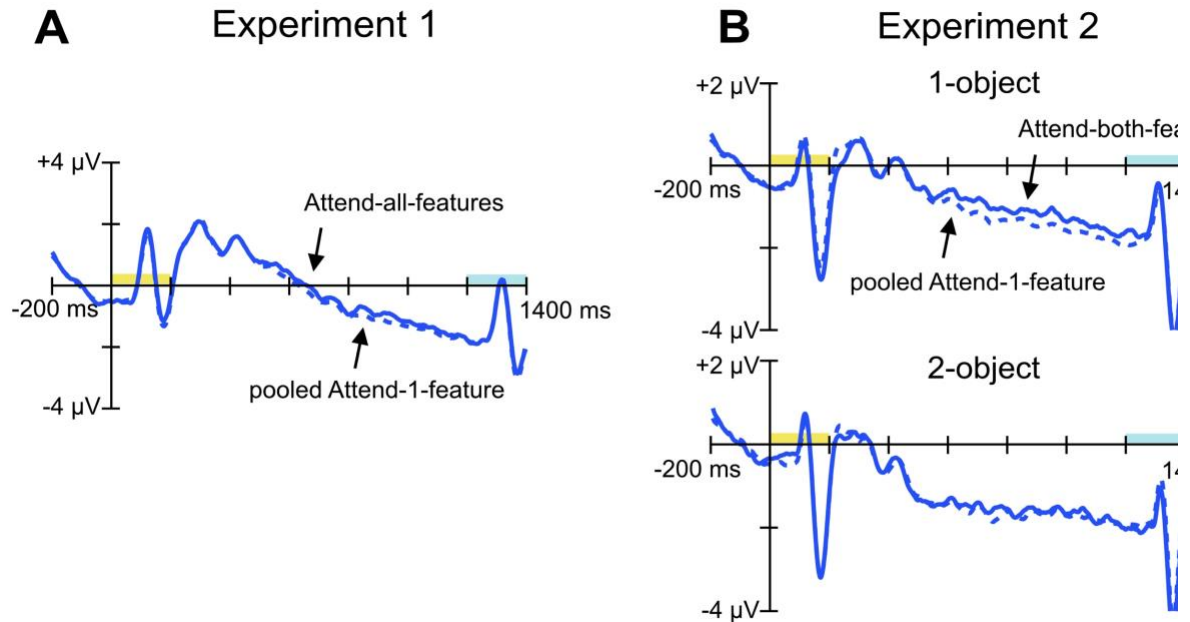
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Negative slow wave (NSW), Contralateral, and Ipsilateral Waveforms

In addition to the lateralized CDA, visual working memory maintenance is associated with a sustained posterior negative slow wave (NSW) that does not depend on contralateral-minus-ipsilateral subtraction. The amplitude of this event-related slow wave becomes increasingly negative as memory load increases and reaches an asymptote near behavioral capacity, suggesting that it reflects the overall amount of information maintained in visual working memory (Fukuda et al., 2015). We therefore conducted an exploratory analysis that examined the NSW to determine whether attending to multiple features elicited greater overall bilateral delay activity than attending to a single feature.

For each participant, the Attend-1-feature conditions were pooled across feature types and compared with the Attend-all-features condition in Experiment 1 or the Attend-both-features condition in Experiment 2 (Figure S1). NSW amplitude was calculated as the mean of the contralateral and ipsilateral amplitudes, $NSW = (\text{contralateral} + \text{ipsilateral})/2$. The measurement window began 200 ms after the offset of the memory array and continued throughout the remainder of the delay period. Bayesian and frequentist tests were conducted in parallel, consistent with the analyses reported in the main text.

Figure S1*Negative Slow Wave Waveforms as a Function of Feature Load in Experiments 1 and 2*

Note. Grand-averaged NSW waveforms are shown time-locked to the onset of the memory array. Panel A shows the pooled Attend-1-feature and Attend-all-features conditions in Experiment 1. Panel B shows the pooled Attend-1-feature and Attend-both-features conditions separately for the 1-object and 2-object conditions in Experiment 2. Solid lines represent the Attend-all-features condition in Experiment 1 and the Attend-both-features condition in Experiment 2; dashed lines represent the pooled Attend-1-feature condition. The yellow bar on the X axis indicates the memory array period from 0 to 200 ms, and the blue bar on the X axis indicates the onset of response period from 1200 to 1400 ms. All waveforms were low-pass filtered at 20 Hz for visualization purposes only; all statistical analyses were conducted without the low-pass filter.

Experiment 1

NSW. NSW amplitude did not reliably differ between the pooled Attend-1-feature condition ($M = -0.55$, $SD = 1.13$) and the Attend-all-features condition ($M = -0.42$, $SD = 1.00$), $BF_{10} = 0.33$, $t(19) = -0.88$, $p = .390$. Thus, attending to all three features did not increase overall bilateral negativity relative to attending to one feature.

Contralateral and Ipsilateral Waveforms. The contralateral waveform was nearly identical across the pooled Attend-1-feature ($M = -1.03$, $SD = 1.21$) and Attend-all-features conditions ($M = -1.01$, $SD = 1.18$), $BF_{10} = 0.23$, $t(19) = -0.10$, $p = .921$. The ipsilateral waveform

was somewhat more positive in the Attend-all-features condition ($M = 0.18$, $SD = 0.98$) than in the pooled Attend-1-feature condition ($M = -0.07$, $SD = 1.13$), but the evidence was inconclusive, $BF_{10} = 0.98$, $t(19) = -1.86$, $p = .079$. These results provide very limited evidence that the feature-load effect observed in the CDA arose from increased contralateral negativity; if anything, the numerical difference was more apparent in the ipsilateral waveform.

Experiment 2

To determine whether the effects observed in the CDA and NSW were associated with changes in contralateral activity, ipsilateral activity, or both, we conducted NSW, contralateral and ipsilateral amplitudes separately to 2×2 repeated-measures ANOVAs with Number of Objects and Number of Attended Features as factors.

NSW. The repeated-measures ANOVA with Number of Objects (1 vs. 2) and Number of Attended Features (pooled Attend-1-feature vs. Attend-both-features) revealed a main effect of Number of Objects, $BF_{10} = 69.40$, $F(1, 23) = 8.84$, $p = .007$. NSW amplitude was more negative for 2-object than 1-object displays. There was no reliable main effect of Number of Attended Features, $BF_{10} = 0.37$, $F(1, 23) = 1.36$, $p = .255$, and no interaction, $BF_{10} = 0.38$, $F(1, 23) = 1.23$, $p = .280$.

Paired comparisons of object load were consistent with this pattern. For pooled Attend-1-feature trials, NSW amplitude was more negative for 2-object ($M = -1.54$, $SD = 1.15$) than 1-object displays ($M = -1.19$, $SD = 0.67$), although the Bayesian evidence was modest, $BF_{10} = 1.34$, $t(23) = 2.09$, $p = .048$. The object-load effect was clearer for Attend-both-features trials (1 object: $M = -0.97$, $SD = 0.96$; 2 objects: $M = -1.49$, $SD = 1.24$), $BF_{10} = 11.62$, $t(23) = 3.25$, $p = .004$. Direct comparisons of pooled Attend-1-feature and Attend-both-features conditions were not

reliable within either 1-object, $BF_{10} = 0.55$, $t(23) = -1.46$, $p = .158$, or 2-object displays, $BF_{10} = 0.23$, $t(23) = -0.44$, $p = .666$.

Contralateral and Ipsilateral Waveforms. For the contralateral waveform, the ANOVA revealed a robust main effect of Number of Objects, $BF_{10} = 5645.48$, $F(1, 23) = 15.87$, $p < .001$, with greater negativity for 2-object than 1-object displays. Neither the main effect of Number of Attended Features, $BF_{10} = 0.22$, $F(1, 23) = 0.13$, $p = .722$, nor the interaction was reliable, $BF_{10} = 0.32$, $F(1, 23) = 0.61$, $p = .443$.

Object-load comparisons confirmed greater contralateral negativity for 2-object than 1-object displays in both the pooled Attend-1-feature condition (1 object: $M = -1.39$, $SD = 0.79$; 2 objects: $M = -1.97$, $SD = 1.25$), $BF_{10} = 10.06$, $t(23) = 3.18$, $p = .004$, and the Attend-both-features condition (1-object condition: $M = -1.29$, $SD = 1.02$; 2-object condition: $M = -1.99$, $SD = 1.30$), $BF_{10} = 51.92$, $t(23) = 3.94$, $p < .001$. In contrast, direct comparisons of pooled Attend-1-feature and Attend-both-features trials showed no contralateral difference within either 1-object, $BF_{10} = 0.26$, $t(23) = -0.68$, $p = .505$, or 2-object displays, $BF_{10} = 0.22$, $t(23) = 0.13$, $p = .895$.

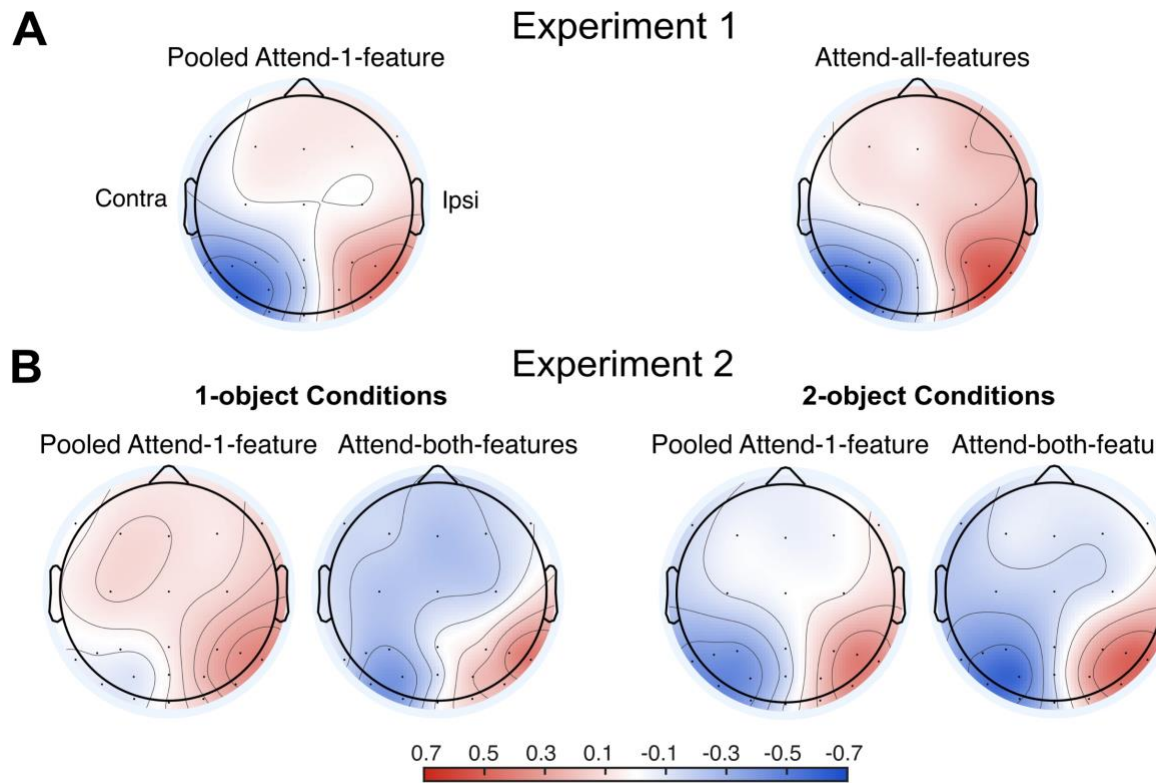
For the ipsilateral waveform, the ANOVA revealed no reliable main effect of Number of Objects, $BF_{10} = 1.09$, $F(1, 23) = 2.42$, $p = .133$, and no reliable main effect of Number of Attended Features, $BF_{10} = 1.24$, $F(1, 23) = 3.33$, $p = .081$. The interaction between Number of Objects and Number of Attended Features was also not reliable, $BF_{10} = 0.35$, $F(1, 23) = 1.67$, $p = .210$. Follow-up comparisons yielded moderate evidence favoring the absence of an object-load effect in the pooled Attend-1-feature condition, $BF_{10} = 0.27$, $t(23) = 0.73$, $p = .473$. In the Attend-both-features condition, the ipsilateral waveform was numerically more negative for 2-object than 1-object displays, but the Bayesian evidence was inconclusive and the frequentist comparison did not reach significance, $BF_{10} = 1.22$, $t(23) = 2.03$, $p = .054$.

We next compared the pooled Attend-1-feature and Attend-both-features conditions within each object-load condition. These conditions did not reliably differ within either 1-object displays (Attend-1-feature: $M = -0.99$, $SD = 0.63$; Attend-both-features: $M = -0.66$, $SD = 0.96$), $BF_{10} = 1.22$, $t(23) = -2.03$, $p = .054$, or 2-object displays (Attend-1-feature: $M = -1.11$, $SD = 1.08$; Attend-both-features: $M = -0.98$, $SD = 1.24$), $BF_{10} = 0.33$, $t(23) = -0.98$, $p = .338$.

Overall, the separate waveform analyses indicate that the robust effect of object load observed in the CDA was primarily associated with increased contralateral negativity. Object load reliably affected the contralateral waveform and overall bilateral activity indexed by the NSW, but did not reliably affect ipsilateral activity. In contrast, the number of attended features produced no reliable modulation of either the contralateral or ipsilateral waveform.

Topographical Distribution of the CDA in Two Experiments

To illustrate the spatial distribution of the CDA, we generated topographical maps of the contralateral-minus-ipsilateral activity for both experiments (Figure S2). The maps were averaged across the complete CDA measurement window (400-1200 ms). In Experiment 1, maps were generated separately for the pooled Attend-1-feature and Attend-all-features conditions. In Experiment 2, maps were generated separately for the pooled Attend-1-feature and Attend-both-features conditions within the 1-object and 2-object trials. Little activity was apparent near the outer canthi, consistent with minimal residual ocular contamination following artifact rejection. Across both experiments and all conditions, the CDA showed a broadly similar posterior distribution, with the strongest activity over occipital-parietal electrode sites.

Figure S2*Topographical distribution of the CDA in Experiments 1 and 2.*

Note. Topographical maps show contralateral-minus-ipsilateral activity averaged across the complete CDA measurement window. All maps use the same voltage scale.

Multivariate Decoding of Object and Feature Load

We also conducted an exploratory analysis on the data from Experiment 2 using multivariate approaches to decode working memory load for EEG scalp activity (Adam et al., 2020; Jones et al., 2024). To quickly summarize our general decoding approach, we used linear support vector machines (SVMs) to classify load conditions from artifact-free, single-trial EEG data (exported in ERPLAB's BEST format). Analyses were conducted within participants using 27 scalp channels and 50-ms windows advanced in 25-ms steps, effectively down-sampling the EEG data to 40 Hz. We used a 5-fold cross-validation procedure. The data was divided into 5 folds: 4 folds served as the training set (80%) and 1 fold served as the held-out test set (20%).

This procedure was repeated 5 times, with each fold serving once as the held-out test set. Trial counts were equated between classes and, in feature-load vs object-load decoding analysis, across the four classifiers. Thus, the decoding results were averaged over 1,000 iterations for each participant. Time-resolved tests focused on the 400-1,200-ms retention interval and were corrected for multiple comparisons using a false discovery rate (FDR) procedure.

Complementary interval-level comparisons used each participant's mean decoding estimate over the same interval.

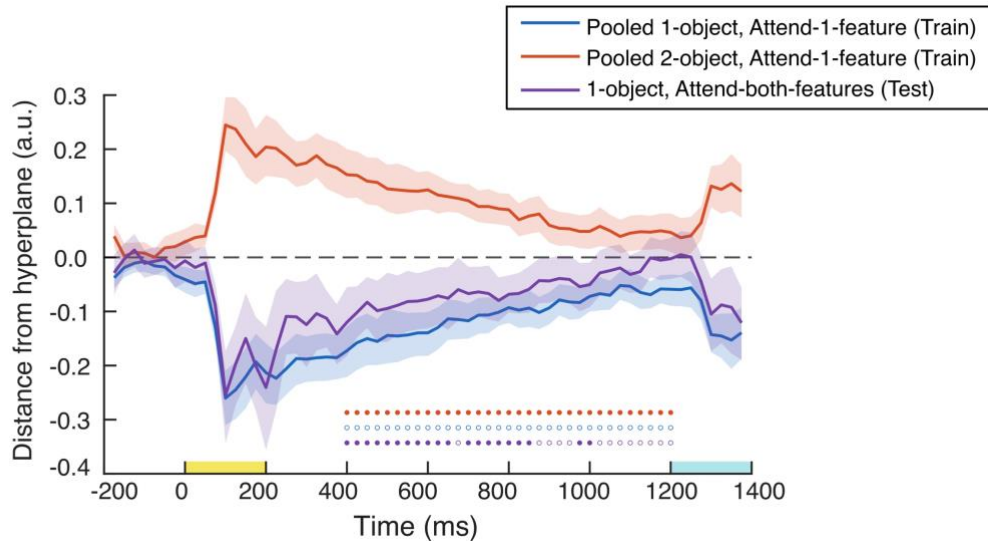
Generalization of Load Decoding

We first asked whether multivariate load decoding generalized according to object load (i.e., the total number of objects) or feature load (i.e., the total number of task-relevant features). The SVM classifier was trained to distinguish between the 1-object, Attend-1-feature condition and the 2-object, pooled Attend-1-feature. As shown in the Figure S3, the classifier was generally successful at distinguishing between the conditions it was trained on. When presented with new

EEG data at test (from these originally trained conditions), the classifier was able to accurately distinguish between the two original conditions (blue vs. red lines).

Figure S3

Multivariate load (mvload) Decoding Results in Experiments 2.



Note. Distance from the classification hyperplane for 1-object, pooled Attend-1-feature condition (blue, a trained condition), 2-object, pooled Attend-1-feature (red, a trained condition), and 1-object, Attend-both-feature condition (purple, the untrained Test condition). The dashed line marks the decision boundary; negative values favor the one-object class. Distances are in arbitrary units (a.u.). Colored circles indicate statistical comparisons of the untrained condition against something else at each tested time point. Red circles, untrained Test condition versus Pooled 2-object, Attend-1-feature; blue circles, untrained Test condition versus Pooled 1-object, Attend-1-feature; purple circles, untrained Test condition versus zero (the classification boundary). Filled circles indicate significance after FDR correction within each comparison; open circles indicate nonsignificant results. Notice that the Test condition did not differ from the other 1-object condition but differed from the 2-object condition. Yellow and blue bars indicate memory array and response onset, respectively.

Importantly, we also tested the classifier on new condition that yields competing predictions: the 1-object, Attend-both-feature condition. This condition has an object load of 1, but a feature load of 2. There are therefore competing predictions. If the trained classifier is primarily object-based, this new condition has the same object load (1) as the 1-Object, Attend-1-feature condition. The neural activity from this new condition should therefore be classified as most similar the blue line. If the trained classifier is instead feature-based, this new condition

requires a feature load of 2, which is the same as the 2-object, Attend-1-feature condition. By this logic, the neural activity from this new condition should therefore be classified as most similar to the red line.

Classification evidence was quantified using signed distance from the decision boundary: Negative values favored the one-object class, and positive values favored the two-object (two-feature) class (see Figure S3). The critical 1-object, Attend-both-features condition fell on the one-object side of the boundary during most of the retention interval. There was decisive Bayesian evidence that its mean distance differed from the 2-object, pooled Attend-1-feature condition, $BF_{10} > 10,000$, $t(23) = -6.43$, $p < .001$. But the comparison with the 1-object, pooled Attend-1-feature condition yielded only weak and inconclusive evidence for a difference, $BF_{10} = 1.85$, despite the frequentist result reaching the conventional significance threshold, $t(23) = 2.28$, $p = .032$. Thus, the Bayesian results indicate that the critical condition (1-object, Attend-both-features condition) was clearly distinguishable from the 2-object condition, however, the evidence was insufficient to determine whether it differed from the 1-object condition. This pattern is compatible with an object-based account but does not provide conclusive evidence that the two one-object conditions were equivalent.

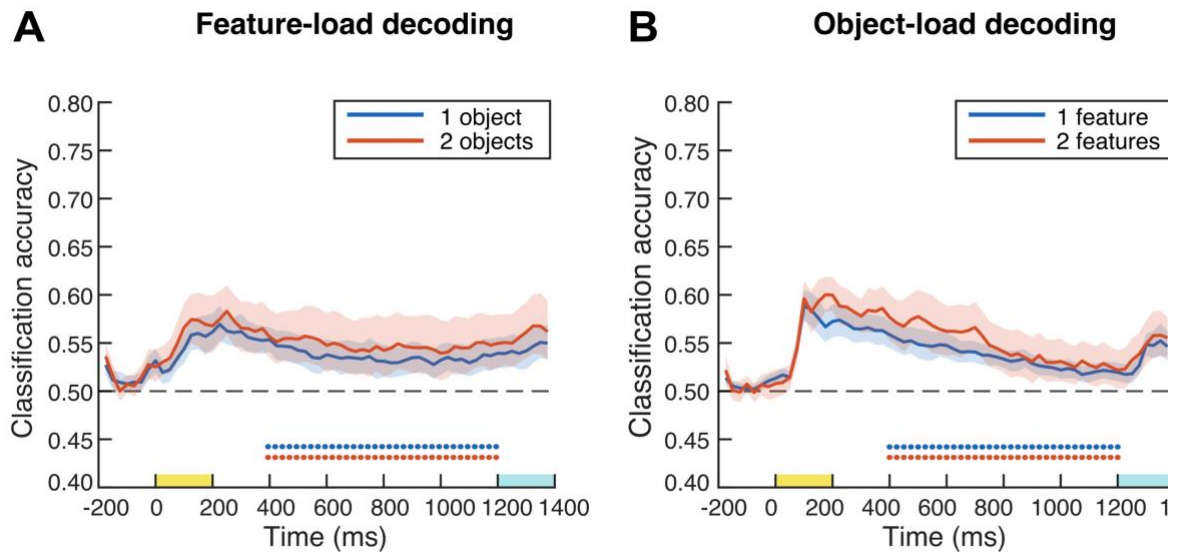
Feature-load vs. Object-load Decoding

We next asked whether feature load and object load could each be decoded when the other factor was held constant. Four independent classifiers were trained to classify feature load (i.e., 1 vs. 2 features per object at each object load), and object load (1 vs. 2 objects at each feature load).

All four classifiers showed above-chance decoding during the CDA time window interval after FDR correction (Figure S4). Mean feature-load decoding accuracy was 53.59% at one object and 54.77% at two objects; mean object-load decoding accuracy was 53.49% at one feature and 54.72% at two features. Averaged across levels of the other factor, feature-load and object-load decoding accuracies did not differ (54.18% vs. 54.11%), $BF_{10} = 0.21$, $t(23) = 0.13$, $p = .898$. In sum, the classification of the 1-object, Attend-both-features condition with the other 1-object conditions is broadly consistent with the CDA's sensitivity to object load, whereas the successful decoding of feature load is consistent with the behavioral evidence that feature load also affected performance.

Figure S4

Multivariate load (mvload) Decoding Results in Experiments 2.



Note. (A) Feature-load decoding at one and two objects. (B) Object-load decoding at one and two attended features per object. The dashed line indicates chance accuracy (.50). Colored dots indicate above-chance decoding after FDR correction. Yellow and blue bars indicate memory array and response onset, respectively.

Altogether, these decoding results suggest there may be both object-based and feature-based representations from the neural activity on the scalp, although it remains to be seen exactly whether this decoding reflects working memory specifically as opposed to some other cognitive process and if so, what exactly about feature-load is represented in working memory. This could be important for future research on this topic.

Relationship Between K Value (Items in Working Memory) and CDA Amplitude

To facilitate comparison with the visual working memory capacity literature, we calculated K , an estimate of the number of items maintained in working memory that accounts for guessing (Cowan, 2001; Rouder et al., 2011). For each participant in a 1-object condition, K was calculated as the $N \times (H - F)$, where N is the number of to-be-remembered objects, H is the proportion of change trials correctly identified as changed, and F is the proportion of no-change trials incorrectly identified as changed. For the 2-object condition, it was calculated as $K = N \times (H - F)/(1 - F)$. This difference in calculation reflects the single-probe situation that Cowan modeled, versus the whole-display probe format that Pashler (1988) modeled (see Cowan et al., 2013, Appendix). We then plotted mean K against mean CDA amplitude across conditions.

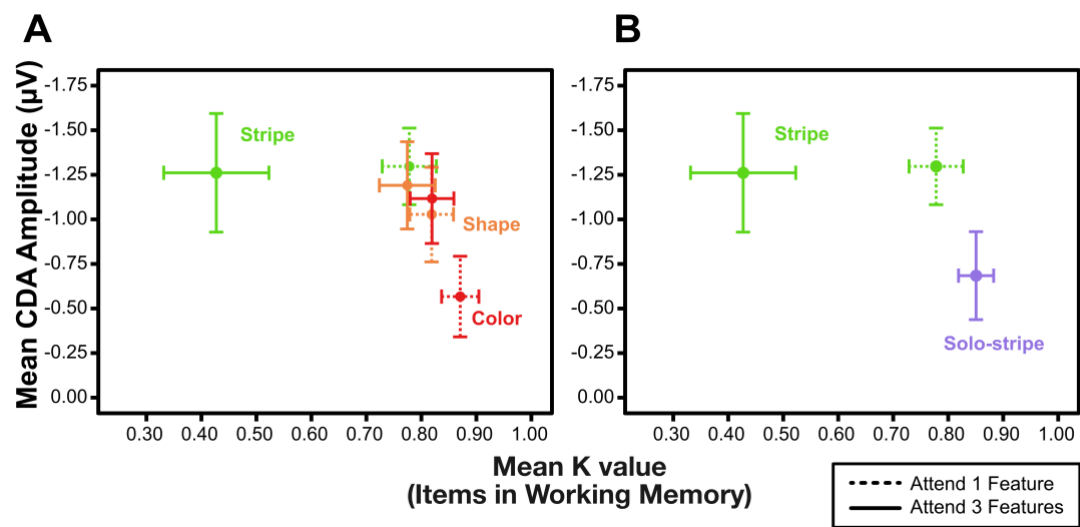
In Experiment 1, mean K was lower when participants attended three features rather than one (0.67 vs. 0.82), with the largest reduction for stripe probes (Table S1). This pattern highly paralleled the accuracy results. CDA amplitude varied relatively little with feature load for shape

and stripe, although it increased for color (Figure S5A). The Attend-stripe condition also yielded lower K but a larger CDA than the Solo-stripe condition (Figure S5B).

Table S1
Mean K value (items in working memory) for each probed feature in the attend-1-feature and attend-All-features conditions in Experiment 1

Probed	Attend-1-features	Attend-all-features
Feature		
Color	0.87(0.09)	0.82(0.10)
Shape	0.82(0.10)	0.77(0.15)

Figure S5
Relation Between K value and CDA Amplitude Across Conditions in Experiment 1



In Experiment 2, mean K was lower when participants attended both features rather than one, with a larger reduction for two objects (1.60 to 1.45) than for one object (0.88 to 0.86; Table S2). These feature-load differences were consistent with the accuracy results. Across object loads, K was higher and CDA amplitude was generally more negative for two objects than for one (Figure S6). Within each object load, CDA amplitude showed little variation with feature load for stripe, with a larger difference for color. In sum, these descriptive patterns suggest that K also captured behavioral costs of increased feature load and object load.

Figure S6
Relation Between K value and CDA Amplitude Across Conditions in Experiment 2

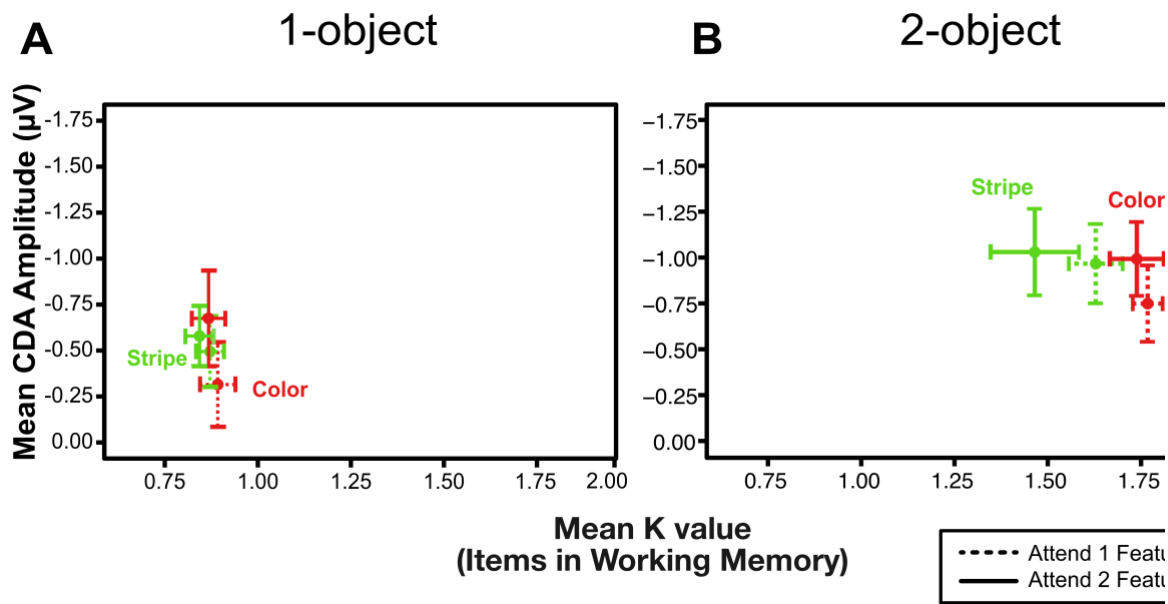


Table S2
Mean K value (items in working memory) for each probed feature in the 1-object and 2-object conditions in Experiment 2

Condition	Probe	Attend-	Attend-
	Feature	both-features	both-features
1-Object	Color	0.89(0.08)	0.87(0.10)
	Stripe	0.87(0.09)	0.84(0.13)
	Both	0.88(0.07)	0.86(0.10)

References in Supplement

- Adam, K. C. S., Vogel, E. K., & Awh, E. (2020). Multivariate analysis reveals a generalizable human electrophysiological signature of working memory load. *Psychophysiology*, 57(12), e13691. <https://doi.org/10.1111/psyp.13691>
- Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24, 87-185. DOI: <https://doi.org/10.1017/S0140525X01003922>
- Cowan, N., Blume, C.L., & Saults, J.S. (2013). Attention to attributes and objects in working memory. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 39, 731-747. <https://doi.org/10.1037/a0029687>
- Fukuda, K., Mance, I., & Vogel, E. K. (2015). α power modulation and event-related slow wave provide dissociable correlates of visual working memory. *Journal of Neuroscience*, 35(41), 14009-14016. <https://doi.org/10.1523/JNEUROSCI.5003-14.2015>
- Jones, H. M., Diaz, G. K., Ngiam, W. X. Q., & Awh, E. (2024). Electroencephalogram decoding reveals distinct processes for directing spatial attention and encoding into working memory. *Psychological Science*, 35(10), 1108-1138. <https://doi.org/10.1177/09567976241263002>
- Rouder, J.N., Morey, R.D., Morey, C.C., & Cowan, N. (2011). How to measure working-memory capacity in the change-detection paradigm. *Psychonomic Bulletin & Review*, 18, 324-330. <https://doi.org/10.3758/s13423-011-0055-3>